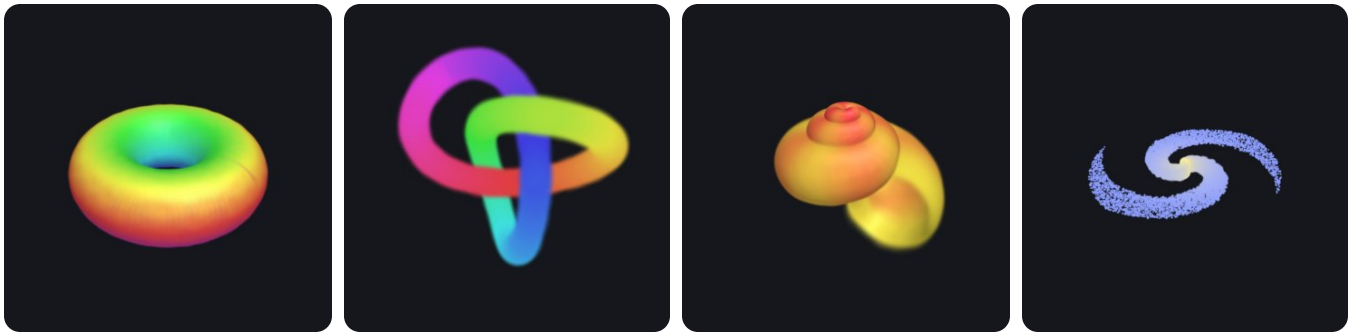
SPLASHERY · UPDATED OCTOBER 11, 2026

The Tinkerer's Manual

How to program Gaussian splats: what one splat is, how the toys are built, and a small language for writing your own splat equations, and toys that show pictures and read what you type.



Every toy in the Splashery Playground is made of Gaussian splats: tens of thousands of soft, colored blobs that blend into a solid-looking thing. This manual is for anyone who knows some math and wants to make splats do what they say. It goes level by level. You can stop after any level and still have something to play with.

One note on that opening line. The photo toys (Photo to 3D and Moving photo to 3D) open in a **Sharp view**, a second way to see the same photo: the picture at its full resolution on a relief lifted by the same depth, so small text stays readable. They are still splats. The Splats / Sharp switch in the toy's panel brings the splats back. (The Studio toys are behind the labs switch; see "What else Splashery does now".)

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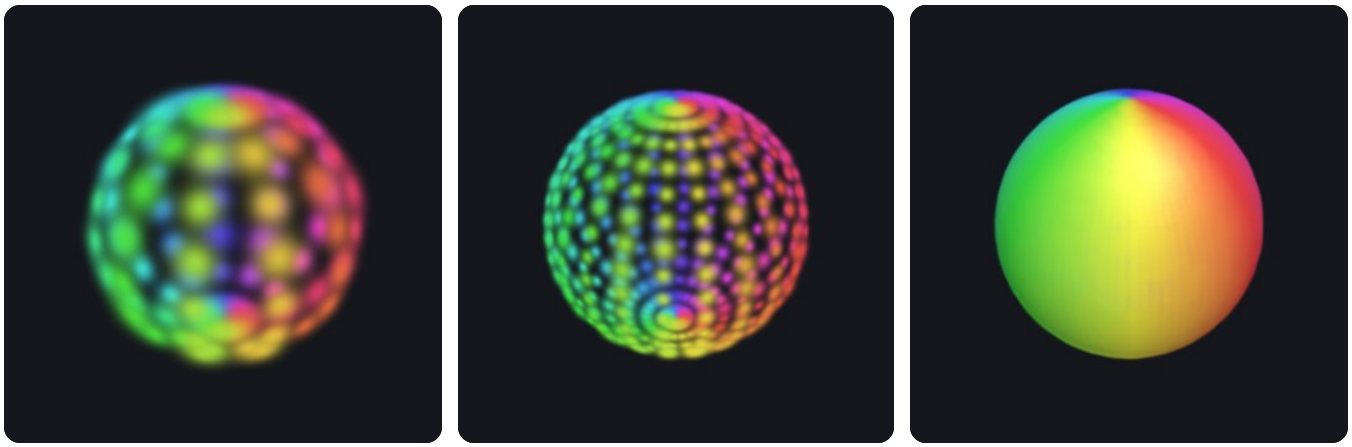
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CHAPTER 1

What a splat is

A **splat** is a small cloud of color floating in space. Up close, it looks like a soft ellipse, bright in the middle and fading at the edges. On its own, it is nothing much. Put thousands side by side, however, and they melt into surfaces: a torus, a seashell, a cat – practically any object imaginable can be built from splats!



100 splats | 400 splats | 6,000 splats

The same sphere with more and more splats, made with the Splat equation toy. With few splats you see each soft blob; with enough of them, they blend into one smooth surface.

What is a 3D Gaussian?

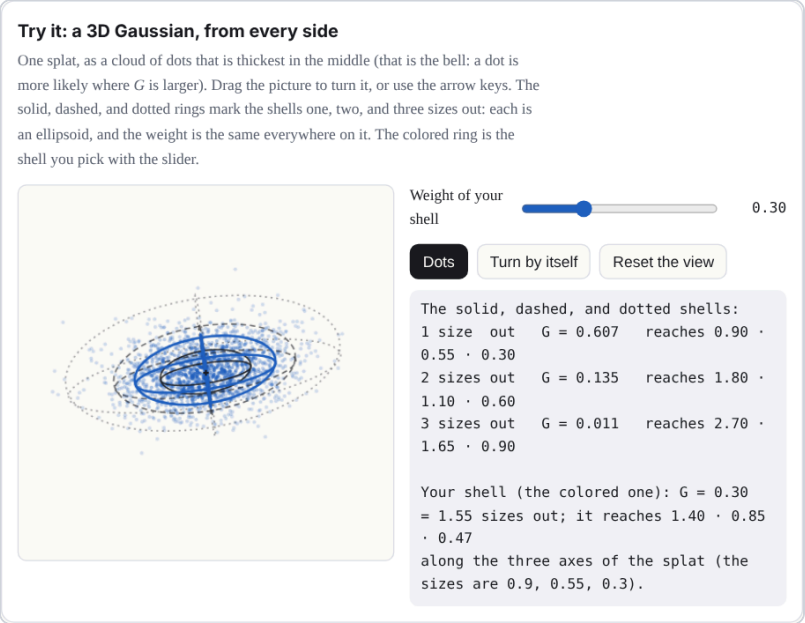
Think of a smooth weight field centered at a point μ. Three positive widths and a rotation shape it: the places where the weight is equal form nested ellipsoids. The weight G is 1 at the center and falls off with distance measured in units of those widths. To draw one, the renderer flattens the field into an elliptical patch on the screen, gives it a color, and blends its opacity over the patches behind it. A splat is neither a surface of triangles nor a rigid little ball. Here G peaks at 1, so it is a weight, not a probability density (which would be scaled to add up to 1).

Two terms: weight field and positive widths

A **weight field** is a number at every point in space: 1 at the center, and falling off the farther you go. The three **positive widths** are the splat's sizes along its own three axes. They must be above zero, because a width of zero would make the formula below divide by zero.

Three ways splats come to exist (Figure 1). Do not mix them up:

- **A recipe's placed splats.** A toy's recipe places them by rule: points sampled on the shapes of **the kit** (Splashery's toy-building library, Level 2), each given a size, a rotation, and a color. Nothing is fitted to photos.
- **A trained capture.** A splat file made by other software from many photos of a real thing, then opened here (`.ply` , `.sog` , `.splat` , or `.spz`). The page shows these; it does not train them.
- **The one-photo teaching fit.** The Gaussian splatting toy fits flat, 2D Gaussians to a single picture by gradient descent. It shows the fitting idea; it is not a 3D reconstruction.



Interactive in the web version: ryanjosephkamp.github.io/splashery/manual/#demo-gaussian

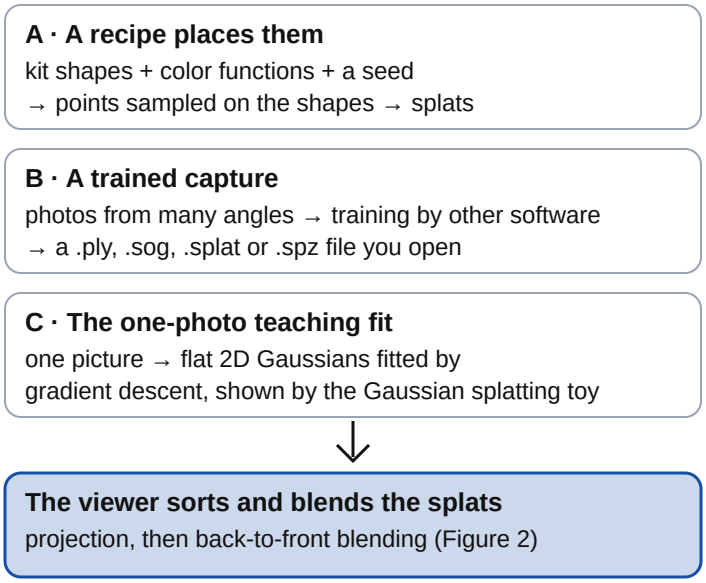


Figure 1. Three different things are called "splats". Only the first two make a 3D object; the third fits a flat picture.

The numbers in one splat

Each splat is described by a handful of numbers:

Part	Numbers	What it means
Center μ	3: x, y, z	Where the middle of the splat is.
Sizes s	3: s_1, s_2, s_3	How far it spreads along each of its own three axes. Equal sizes make a ball; one small size makes a flat disc.
Rotation q	4 (a quaternion)	Which way those three axes point.
Color	3: red, green, blue	Each from 0 to 1.
Opacity α	1	How solid its middle is, from 0 (clear) to 1 (solid).



Interactive in the web version: ryanjosephkamp.github.io/splashery/manual/#demo-numbers

That is 14 numbers per splat. A kit toy starts with a budget of 60,000 splats at the Low tier, 140,000 at Mid, 200,000 at High, and 280,000 at Max. Recipes can raise

that budget, up to 400,000 at Max. Auto picks Low, Mid, or High; Max is a viewer choice. Many splat files also let a splat's color change a little with the direction you look from; the toys you build here use one color per splat.

Why these numbers? The splat budgets, and what Auto does

OUR CHOICE Nothing in the splat file format or the browser sets these four numbers.

Each tier is a splat count that a device of that class can draw smoothly. **Auto** reads the device to pick one: Low for 2 GB of memory or less, or two cores or fewer; Mid for a touch-screen device whose shorter side is under 820 pixels, or 4 GB of memory or less, or four cores or fewer; High for the rest. It never picks Max. That is a choice you make in Detail, along with High. Auto also steps down while you play: when frames take longer than about 24 milliseconds, it first lowers the screen resolution, and if frames at the lowest resolution still average over 40 milliseconds for 30 frames in a row, it drops one tier. The project's notes call the tier thresholds and the 24 and 40 millisecond limits "guesses, not tuned on real phones".

What the measurements say. The Sharpness lane (September 29, 2026) measured 14 toys on a 390 by 844 phone in a software renderer. A budget of 200,000 splats instead of 140,000 made no toy sharper, and several shimmered more (the penguin's shimmer went from 2.72 to 3.90), because each splat got smaller. Lifting the cap on the screen's pixel ratio was the real lever: edges came out about a third narrower, and speckle fell by 10 to 60 percent. More splats help a toy's shape, not its grain.

Show the table: what a bigger budget did

Toy (home view)	Today: 140,000 splats, 2x phone	High tier: 200,000 splats	3x phone, pixel-ratio cap lifted
Lamp	0.75 · 0.42 · 1.95	0.72 · 0.45 · 2.03	0.52 · 0.37 · 1.57
Penguin	0.72 · 0.53 · 2.72	0.73 · 0.56 · **3.90**	0.50 · 0.40 · 2.08
Klein bottle	0.84 · 1.19 · 1.42	0.81 · 1.00 · 1.75	0.53 · 0.77 · 1.13
Marble	0.81 · 0.69 · 1.13	0.78 · 0.83 · 1.41	0.55 · 0.48 · 0.87
White blood cell	0.77 · 1.28 · 1.64	0.77 · 1.39 · 1.02	0.54 · 0.75 · 1.15
Chladni plate	0.96 · 3.85 · 9.59	0.98 · 5.51 · 10.75	0.75 · 1.48 · 5.39

Each cell is edge · speckle · shimmer, and lower is better for all three. Edge is how many screen pixels an edge takes to go from 10 to 90 percent; speckle is the noise over flat areas; shimmer is how much fine detail flickers frame to frame. Six of the 14 toys measured; the rest are in `docs/lab/SHARPNESS.md`.

What it did not test. One pair of budgets (140,000 and 200,000), one phone size, one renderer (software, not a real phone's graphics chip), and the toys as they were on that day, before later changes. The Kit lab's larger sweep is in Level 2, under How many splats?.

If it were chosen differently. A bigger budget costs memory and drawing time on a weak device. A smaller one makes shapes coarser. With labs on, a Detail slider now sets the count directly, from 60,000 to 400,000, beside the tiers (`docs/lab/DETAIL-SLIDER.md`); the tiers themselves have not changed.

Where it lives: `PROFILES` in `src/generators.js` ; `detectProfile` in `src/player.js` ; `timeFrame` in `src/stage.js` ; the measurements in `docs/lab/SHARPNESS.md` .

Why this number? One color per splat

OUR CHOICE A captured file can carry more; the kit does not.

A captured file can store how each splat's color changes with the direction you look from (spherical harmonics), because training sees every splat from many angles. A kit splat is placed by a recipe, so there are no photos to learn that change from. The kit keeps one color per splat, and the harmonics of a trained file are dropped when it is used as a kit part.

If it were chosen differently. A recipe can already fake two such looks: `glint` brightens a splat as the camera moves, and `rim` changes glass's opacity with the view. A real stored shine, with the color rule coming from the recipe, is a planned experiment in the Kit lab. It would use the kit's own data and shader, with no change to the engine.

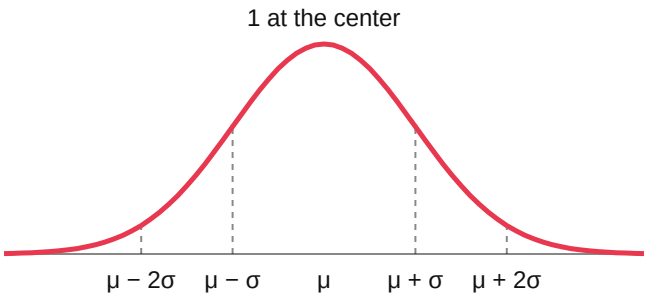
Where it lives: `src/packs/fidelity.js` (its opening comment), `src/kit.js` , and the effect list in `src/effects.js` .

The Gaussian: a bell in three dimensions

The "Gaussian" in Gaussian splat is the bell curve, named after the mathematician Carl Friedrich Gauss. In one data dimension, a bell centered at μ with width σ (sigma, the standard deviation) is

$$G(x) = \exp(-(x - \mu)^2 / 2\sigma^2)$$

The data here are one-dimensional: one number, x, goes in, and one number, G, comes out. The drawing below needs two dimensions to show them, one for x and one for G.

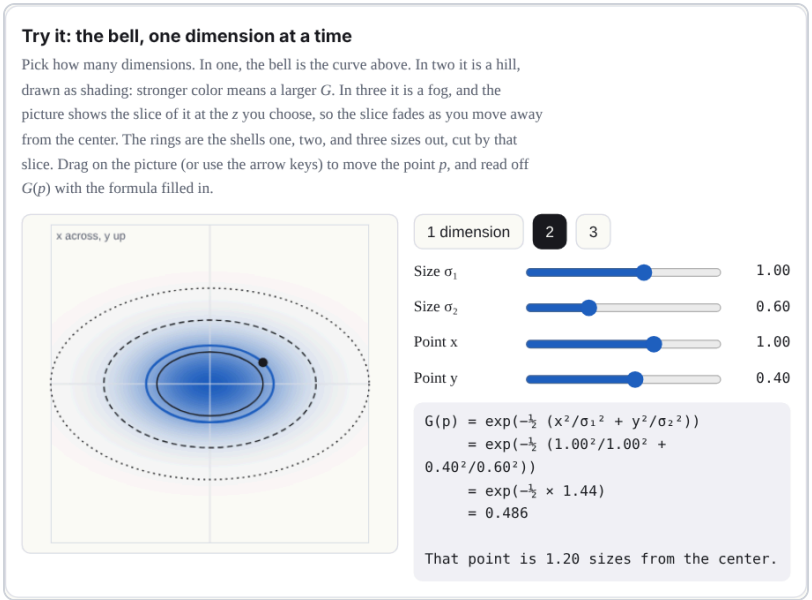


The bell curve. At one σ from the center it has fallen to about 0.61, at two σ to about 0.14, and at three σ to about 0.01, so a splat fades out within about three of its sizes.

In three dimensions, the width becomes a 3-by-3 matrix Σ (capital sigma), called the **covariance** [4], and the squared distance becomes a quadratic form. For a point p in space, the splat's strength there is

$$G(p) = \exp(-\tfrac{1}{2}\,(p-\mu)^{\mathrm{T}}\,\Sigma^{-1}\,(p-\mu))$$

Where G is the same value, you are on the same "shell" around the center, and those shells are ellipsoids: stretched and turned spheres.



Interactive in the web version: ryanjosephkamp.github.io/splashery/manual/#demo-bell

Two words in that formula: the inverse and the quadratic form

The inverse Σ^{-1} is the matrix that undoes Σ. And a **quadratic form** is an expression where every term has two factors of the input, like $ad_1^2 + 2bd_1d_2 + cd_2^2$. In two dimensions, with $d = (d_1, d_2) = p - \mu$, and Σ^{-1} with 4 and 1 on its diagonal (and zeros elsewhere), the form $d^{\mathrm{T}}\,\Sigma^{-1}\,d$ is $4d_1^2 + d_2^2$. Setting it equal to 1 draws an ellipse that reaches ½ along the first axis and 1 along the second. A quadratic form is what turns a point's offset from the center into one number for how far out it is, measured in the splat's own widths. In three dimensions, the same idea draws an ellipsoid [5].

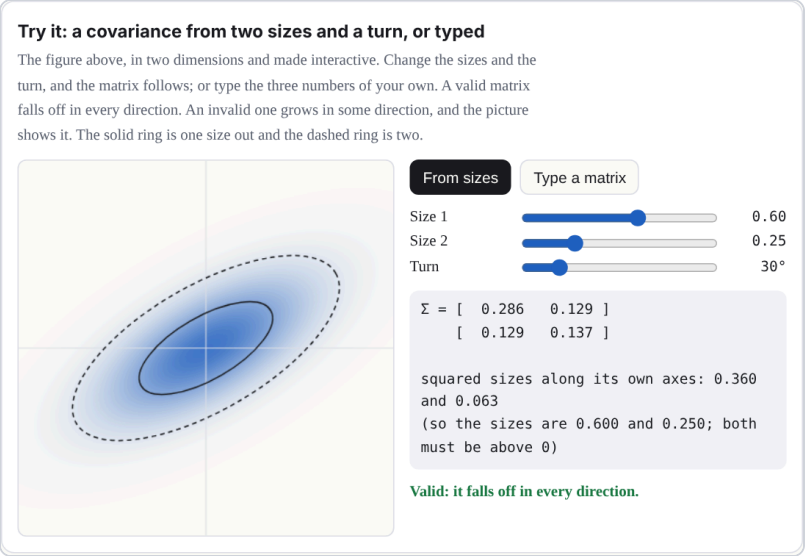
Approximation: the drawn bell is finite. The ideal G never reaches zero. The site's renderer cuts each splat off where G has fallen to e^{-4}, about 0.02 (a little under 3 sizes out), and rescales the falloff so it reaches exactly 0 there. A labs-only setting can swap in a sharper falloff for one toy. So the equations here describe the ideal splat, and the screen shows a close, cut-off copy of it.

The covariance, built from sizes and a rotation

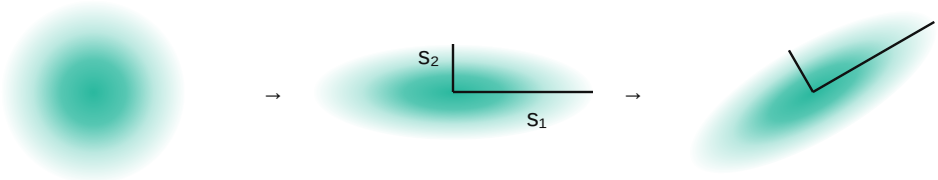
Not every 3-by-3 matrix is a valid covariance. It has to be symmetric, and it has to be positive definite (the bell must fall off in every direction). Instead of picking its nine numbers directly, a splat builds it from its three sizes and its rotation:

$$\Sigma = R\,S\,S^{\mathrm{T}}\,R^{\mathrm{T}}, \quad \text{where} \quad S = \begin{pmatrix} s_1 & 0 & 0 \\ 0 & s_2 & 0 \\ 0 & 0 & s_3 \end{pmatrix}$$

and R is the rotation matrix of the quaternion q. Read it right to left, as a recipe for a shape: start with a round ball of size 1, stretch it by s_1, s_2, and s_3 along x, y, and z, then turn it by R. Any sizes above zero and any rotation give a valid covariance, which is exactly what you want when a computer is nudging millions of numbers.



Interactive in the web version: ryanjosephkamp.github.io/splashery/manual/#demo-covariance



A covariance from sizes and a rotation, drawn in two dimensions: a round splat is stretched along its own axes, then turned.

A worked example: a covariance from sizes 0.3, 0.1, and 0.1

Take sizes 0.3, 0.1 and 0.1, turned 30° about the z axis. Then SS^{T} has 0.09, 0.01, and 0.01 on its diagonal, and multiplying out $R\,SS^{\mathrm{T}}\,R^{\mathrm{T}}$ gives

$$\Sigma \approx \begin{pmatrix} 0.070 & 0.035 & 0 \\ 0.035 & 0.030 & 0 \\ 0 & 0 & 0.010 \end{pmatrix}$$

The top-left number is $0.09\cos^2 30° + 0.01\sin^2 30° = 0.07$, and the ones beside it are $(0.09 - 0.01)\sin 30°\cos 30° \approx 0.035$. Those off-diagonal numbers are the **tilt**: they are zero when the splat's axes line up with x, y, and z, and not zero when they do not.

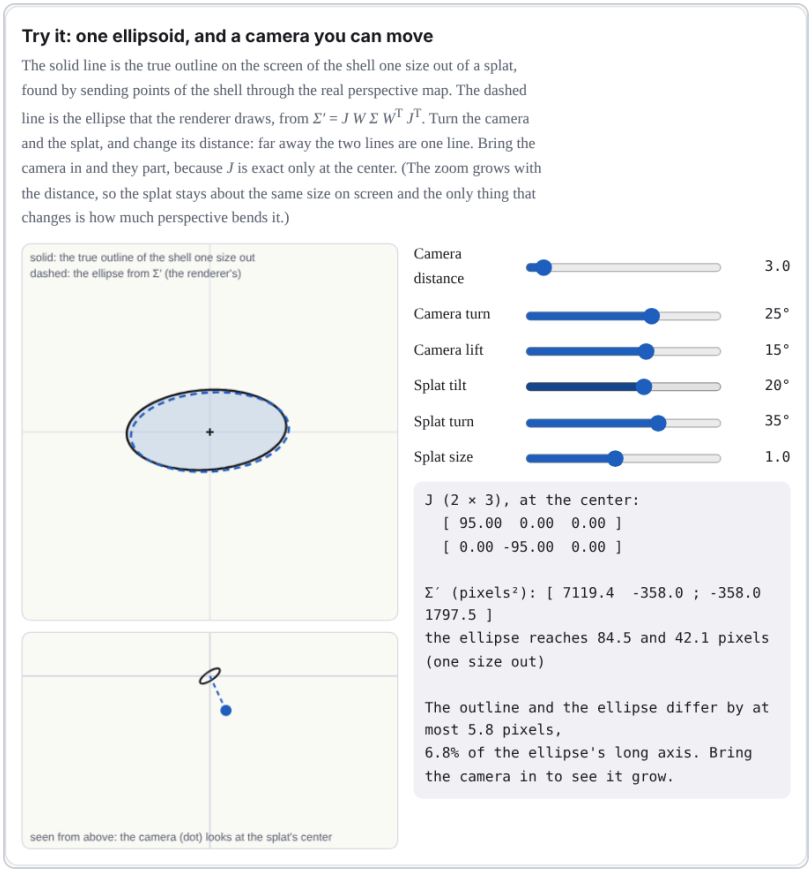
> (In this manual, **rotation** means a splat's own turn, and "Tilt lock" is the name of the camera control in Level 5.)

From 3D to the screen

To draw a splat, the renderer works out what its bell looks like from the camera. It first turns the covariance into the camera's view, with the camera's rotation W. The perspective of a camera is not a straight-line map, but close to one point it nearly is, and its best straight-line stand-in there is a matrix J (the Jacobian of the projection). Such a stand-in is called **affine**: a straight-line map plus a shift [6]. Then the splat's covariance on the screen is

$$\Sigma' = J\,W\,\Sigma\,W^{\mathrm{T}}\,J^{\mathrm{T}}$$

keeping only its top-left 2-by-2 block. That 2-by-2 matrix is a flat bell on the screen: an ellipse. Its eigenvectors are the directions of the ellipse's axes, and the square roots of its eigenvalues are how far it spreads along them. This trick comes from EWA splatting (Zwicker, Pfister, van Baar, and Gross, 2001) [2].



Interactive in the web version: ryanjosephkamp.github.io/splashery/manual/#demo-projection

> **Approximation: affine around the center.** An affine map is a straight-line map plus a shift. J is exact only at the splat's center, where the straight-line stand-in touches the true perspective map. Across the rest of the splat the real map bends away from it, so the ellipse is the first-order answer, not an exact image of the 3D bell.

Sorting and blending, back to front

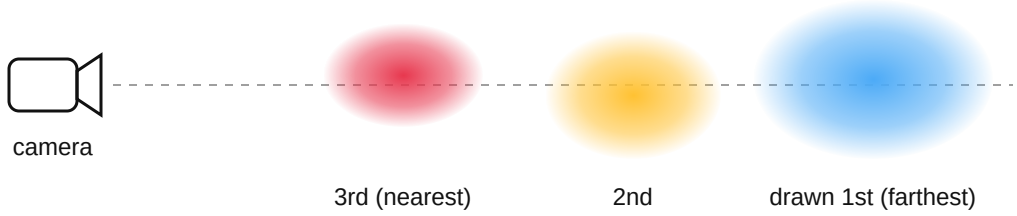
Splats are see-through at their edges, so the order they are drawn in matters. The splats are sorted by their centers' depth along the camera's view, and painted from the farthest to the nearest. At each pixel, a splat's strength there is its opacity times its bell, $a = \alpha\, G'$, and it is laid over what is already there:

$$C \leftarrow a\, c + (1 - a)\, C$$

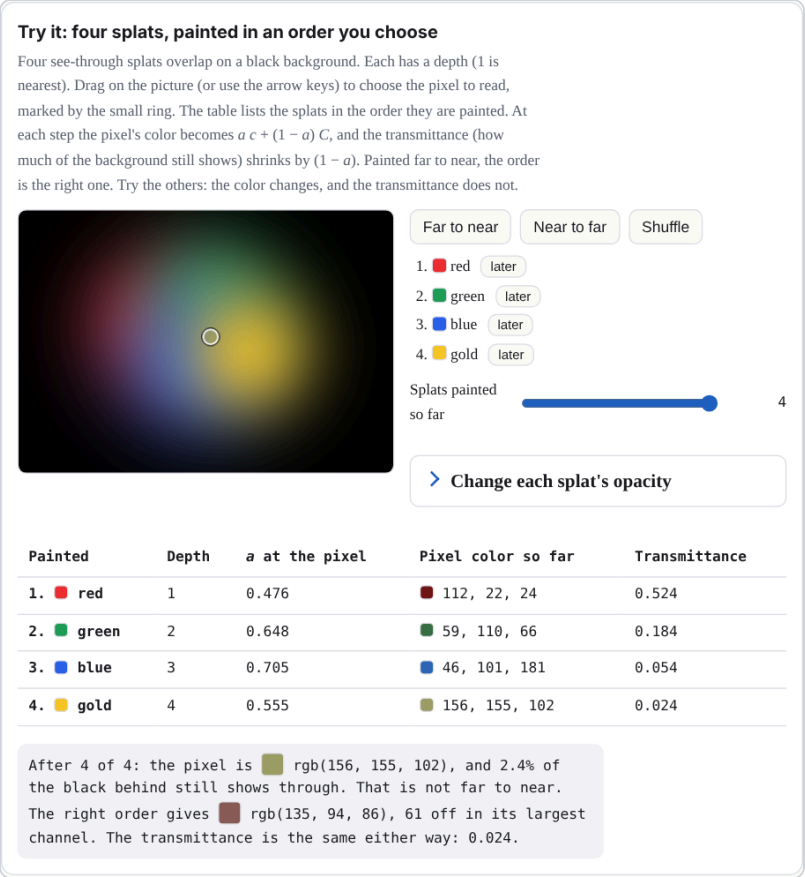
where c is the splat's color and C the pixel's color so far. Written out for all the splats at once, counting from the nearest ($i = 1$) to the farthest, the pixel's color over a black background is

$$C = \Sigma_i\, c_i\, a_i\, \Pi_{j<i}\, (1 - a_j)$$

Each splat shows as much as the splats in front of it let through. This "over" operation is from computer graphics' classic paper on compositing [3]. It is how the 2023 paper that started the craze, "3D Gaussian Splatting for Real-Time Radiance Field Rendering" (Kerbl, Kopanas, Leimkühler, and Drettakis) [1], draws its scenes, and it is how Splashery draws every toy.



Splats are painted from the back to the front, each one laid over the ones behind it.



Interactive in the web version: ryanjosephkamp.github.io/splashery/manual/#demo-blend

Why the order matters in the toys. By default, Splashery sorts the splats in the pose a toy was built in, so a piece that moves keeps the order it had at rest. A recipe can request `out.resortPose` to sort moving parts and pages in their current pose. Turn a solid thing more than a quarter turn and its far side can be drawn over its near side. The toys work around it: the Splat equation toy, for one, builds its shape at twelve moments and morphs from the copy at the start of the current interval to the next (Level 3).

A worked example of blending, with a background

Take one color channel. The background is 0.2. The far splat (drawn first) has color 0.9 and strength $a = 0.5$ at this pixel. The near splat (drawn second) has color 0.1 and $a = 0.4$. After the far splat: $C = 0.5 \times 0.9 + 0.5 \times 0.2 = 0.55$. After the near one: $C = 0.4 \times 0.1 + 0.6 \times 0.55 = 0.37$. The same number comes from the sum above with the background added: $0.1 \times 0.4 + 0.9 \times 0.5 \times 0.6 + 0.2 \times 0.6 \times 0.5 = 0.37$. The share of the background that still shows, the **transmittance**, is $0.6 \times 0.5 = 0.3$.

Approximation: sorting uses centers. Each splat gets one depth, the depth of its center, so the order is per splat, not per pixel. Two big splats that overlap in depth can be drawn in the wrong order where they cross. This is why recipe authors keep layers apart (the recipe guide says splats "sort by their centers, so closer layers mix in stripes").

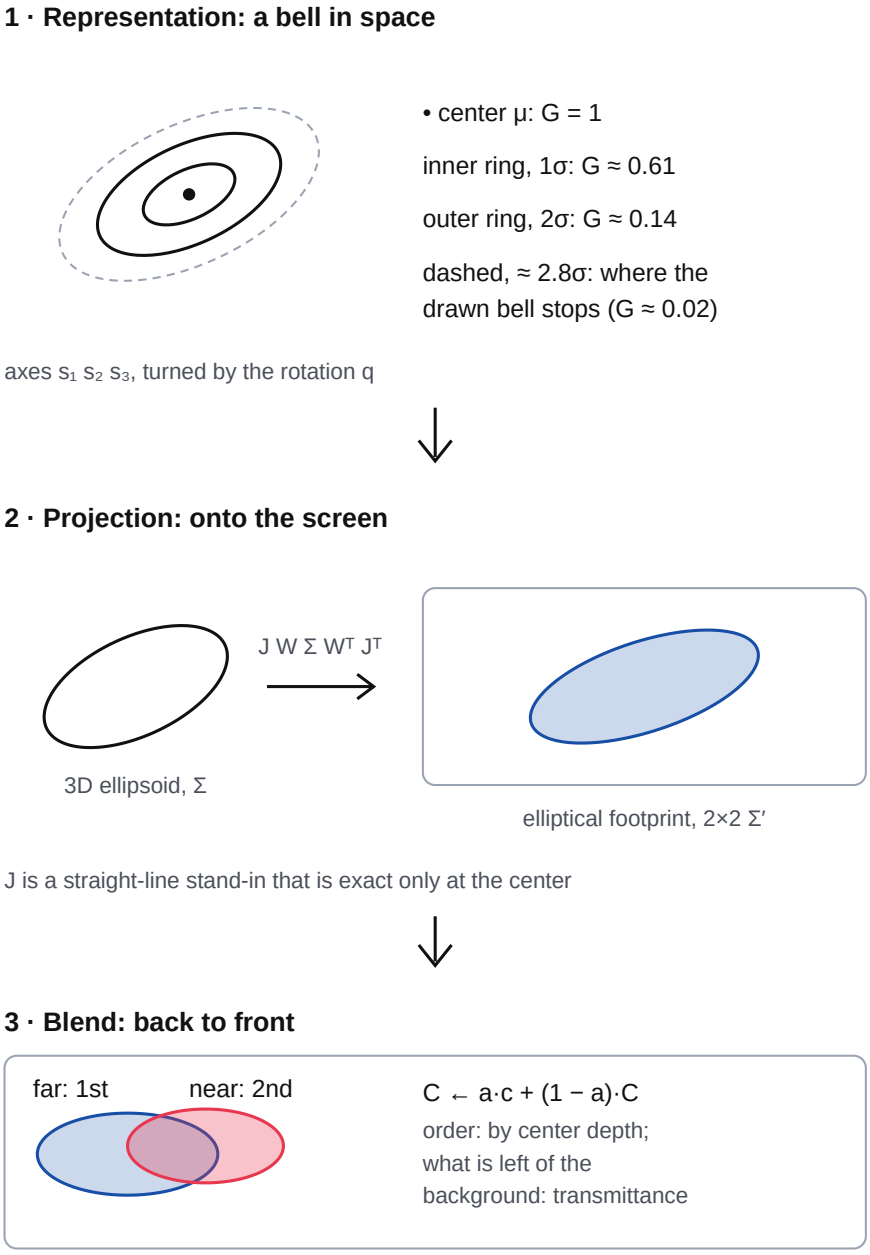


Figure 2. From a splat to a pixel. **1** The ideal bell has equal-weight shells; the drawn one is cut off. **2** The camera flattens the ellipsoid into an ellipse, using a straight-line stand-in for the perspective. **3** The ellipses are blended back to front, each over what is already there.

CHAPTER 2

How a toy is built

Nobody places Splashery's splats by hand. Most toys are **recipes**: short programs that describe a toy with a kit of shapes, and the kit fills them with splats when you open the toy. A recipe is a few kilobytes of code instead of megabytes of splats, and because every random choice comes from a seed, it builds the same splats with the same seed, options, and budget.

The **kit** is Splashery's toy-building library. A recipe names shapes, colors, and moving parts; the kit fills them with splats in the browser, spreading the budget evenly by area. A **kit toy** is built this way. A scan or capture is loaded from a file instead.

Recipes are written in JavaScript. This level is a tour, so you can read one; the complete guide is docs/PACKS.md in the source code [10].

Two places a toy lives

A toy's name and shelf go in a **catalog** entry (`src/toys.js`), and the toy itself in a **recipe** in a pack file (`src/packs/…js`):

```
{
  id: "little-windmill",       // lower-case words joined by hyphens
  label: "Little windmill",    // its name on the shelf
  category: "objects",         // which shelf
  kind: "kit",
  pack: "my-pack",             // loads src/packs/my-pack.js
}
```

A recipe has six basic parts, all optional except `build` (Level 5 adds `pictures` , `input` , and `tiltLock` , and PACKS.md has more, such as `grab` and `drag` for touch play):

Show the table: the six parts of a recipe

`build(k, o)`	Describes the toy with the kit `k` . `o` holds the chosen options.
`options`	Choices that rebuild the toy: a color, a style, a number of petals. They are saved in links.
`controls`	Live values from 0 to 1: sliders, toggles that ease on and off, and pulses that jump to 1 and fall back.

`action`	Which control a tap on the toy fires.
`drive(t, c, out)`	Runs every frame: from the time `t` and the controls `c`, it sets how the parts move.
`alive`	`true` when the toy moves by itself, so frames keep coming.

Shapes from the kit

`k.sphere(r)`, `k.ellipsoid(a, b, c)`, `k.box(w, h, d)`, `k.roundedBox(w, h, d)`, `k.cylinder(r, h)`, `k.cone(r0, r1, h)`, `k.torus(R, r)`, `k.disc(r)`, `k.lathe(profile)` (a vase or a bottle, turned around an axis), `k.tube(curve, radius)` (a stem, an arm, a knot), `k.param((u, v) => [x, y, z])`, any surface you can write in two variables (the same idea as Level 3; here *u* and *v* each run from 0 to 1), and `k.radial((dir) => r)`, a blobby surface given by its distance from the middle in every direction. Nothing is drawn until you add a shape:

```
k.add(k.sphere(0.2), { pos: [0, 1, 0], color: "#ffcc33", weight: 2 });
```

The kit shares the splats out between the shapes by their surface area, so the toy is evenly dense; `weight` gives a small shape more (or `share` a fixed fraction of all the splats). For free-form splats, like stars, sparks, and crumbs, `k.cloud` places them one by one. After `build`, the kit centers the toy and scales it to fit the view, so you can model in any units.

How many splats? What a bigger budget does

The kit spreads a budget of splats over a toy's surface (see Level 1 for the four numbers). It is natural to think more splats always look better. The Kit lab measured it (`docs/lab/BUDGETS.md`, October 10, 2026). It drew the same toy from the same view on the same simulated 390 by 844 phone at five budgets, changing only the count. The table compares 140,000 (the Mid tier) with 200,000 (the High tier) for 24 views of 22 toys.

What the columns mean

- **Edge** is how many screen pixels the strongest edges take to go from 10 to 90 percent of the way between the two colors. A sharp edge is narrow, so lower is sharper.

- **Speckle** is how far pixels on the flatter half of the toy differ from the middle of their neighbors: grit on a smooth surface. Lower is cleaner. Painted grain, like the oak's bark, also counts, because it looks the same at phone size.
- **Shimmer** is how much the fine detail changes from one frame to the next while the toy turns a little. Lower is steadier.
- **Area** is how much of the screen the toy covers, as a percent of the same view at 400,000 splats. Over 100 percent means thin parts drawn fatter than they are. The table shows 60,000 and then 140,000 splats, because that is where the change is.

Same toy, same view, same phone: 140,000 splats → 200,000 splats. Lower is better in every column. Worse Better The same On a screen, the table scrolls sideways inside its box.

Toy (view)	Edge (screen pixels)	Speckle	Shimmer	Area, 60k → 140k
Lamp	0.52 → 0.51	**0.37 → 0.39**	**1.57 → 1.65**	101% → 100%
American football	0.50 → 0.48	**0.61 → 0.70**	**1.49 → 1.60**	100% → 100%
Hockey puck	**0.27 → 0.29**	0.24 → 0.21	1.01 → 0.99	102% → 101%
Klein bottle	0.56 → 0.53	**0.52 → 0.90**	**1.12 → 1.33**	100% → 99%
Clock	0.51 → 0.49	**0.49 → 0.54**	**1.64 → 1.90**	101% → 100%
Penguin	0.51 → 0.49	**0.42 → 0.46**	**2.92 → 3.13**	–
Snow globe	0.44 → 0.44	**0.20 → 0.23**	**0.71 → 0.84**	102% → 101%
White blood cell	0.54 → 0.53	**0.72 → 0.82**	**0.84 → 1.02**	101% → 100%
Ocean liner	0.46 → 0.44	**0.31 → 0.35**	**1.69 → 2.09**	106% → 104%
Marble	0.46 → 0.45	**1.43 → 1.65**	**2.63 → 2.96**	100% → 100%
Sailboat	0.50 → 0.46	**0.51 → 0.59**	**2.26 → 2.93**	103% → 103%
Toy piano	0.60 → 0.56	**0.22 → 0.25**	**1.93 → 2.32**	111% → 105%
Your book	**0.52 → 0.55**	0.00 → 0.00	1.17 → 0.46	–
Chladni plate	0.66 → 0.66	0.26 → 0.26	6.02 → 6.01	100% → 100%
Bicycle	0.50 → 0.49	**0.89 → 0.99**	**6.00 → 6.52**	118% → 107%
Eiffel tower	0.58 → 0.55	**0.20 → 0.24**	**0.86 → 1.19**	109% → 104%
Crystal lattice	0.57 → 0.54	**0.27 → 0.31**	**1.49 → 1.83**	111% → 105%
Knot	0.53 → 0.52	**0.48 → 0.58**	**1.54 → 1.87**	103% → 101%
Torus knot	0.53 → 0.52	**0.29 → 0.34**	**1.10 → 1.25**	98% → 98%

Toy (view)	Edge (screen pixels)	Speckle	Shimmer	Area, 60k → 140k
Oak	0.58 → 0.57	**0.40 → 0.51**	**0.82 → 1.03**	105% → 102%
Chess set	0.58 → 0.52	**0.14 → 0.15**	**2.44 → 3.09**	106% → 103%
Dna	0.51 → 0.50	**0.50 → 0.57**	**4.06 → 4.53**	–
Bicycle (close up, 0.5)	0.53 → 0.52	**0.36 → 0.42**	**2.78 → 3.25**	118% → 107%
Chess set (close up, 0.6)	0.60 → 0.54	**0.08 → 0.09**	**1.50 → 2.01**	103% → 101%

All five budgets across all the views: the middle value of each toy's change against its own result at 140,000 splats.

Budget	Edge	Speckle	Shimmer	Thin parts' area
60k	+11%	−23%	−36%	+1.2%
140k	–	–	–	–
200k	−2%	+14%	+18%	−0.8%
280k	−6%	+31%	+39%	−0.8%
400k	−9%	+52%	+61%	−1.2%

What it shows

- **Going from 140,000 to 200,000, speckle rose in 21 of the 24 views and shimmer rose in 21 of them** (the counts are from the table). Across all the views the middle change was 14 percent more speckle and 18 percent more shimmer. Each splat gets smaller as the count goes up, so the fine detail in a pixel comes from more, smaller splats that move in and out of it as the view turns.
- **Edges barely changed.** The most any edge gained between the two budgets was 0.06 of a screen pixel, the middle change was 2 percent, and in two views the edge got a little softer.
- **The one place a bigger budget clearly helps is thin parts.** At 60,000 splats the bicycle's frame and spokes cover 18 percent more of the screen than at 400,000; at 140,000 they are within 7 percent. Solid toys, like the clock or the marble, stay within 1.5 percent of the 400,000 view even at 60,000 splats.
- **No holes open at the lower budgets.** A low budget makes thin parts fatter, not broken.

What it does not show: one renderer (a software one, not a real phone's graphics chip), one phone size, still views with a slow turn, and no frame rate you could trust. Scans (captured toys) are not affected, because they load their own files. The kit

lab's recommendation was to keep the four tiers as they are. The full tables, the method, and the limits are in docs/lab/BUDGETS.md and the first round of this question is in `docs/lab/SHARPNESS.md`.

Colors from functions

A color can be a fixed quantity or vector (`"#9c3b2c"`, or `[r, g, b]` with each from 0 to 1; there are no color names), or it can be a function that the kit calls for every splat. The function gets the splat's place `c.p`, its surface direction `c.n`, its surface coordinates `c.u` and `c.v`, and a random number and noise that come from the seed. Return `null` to leave a hole:

```
color: (c) => (c.p[1] > 0.5 ? "#ffffff" : "#3a7d44"), // snow above, grass below
```

Parts that hinge, spin or slide

A **part** is a group of shapes that moves as one solid piece: `k.part("lid", { pivot, axis })`. Every frame, `drive` tells each part where to be: an `angle` about its axis (a lid on a hinge, a wheel that spins), an `offset` (a drawer that slides), a `scale`, or `visible`. A toy may have up to 15 parts. The treasure chest's lid is a part on a hinge.

> **Why this number? Fifteen parts**
>
> OUR CHOICE, NEAR A LIMIT The format of the part number sets the count; the shader's space makes it hard to raise.
>
> Every splat carries the number of the part it belongs to, and that number gets only 4 bits of a packed value, because the same value also holds two flags for the shader. Four bits count 16 things. Number 0 is the body, which every toy has, so 15 parts are left to name. Asking for a sixteenth stops the build with a message. The shader keeps three vector values for each of the 16 parts (a rotation, a pivot, and an offset with visibility), 48 in all.
>
> **If it were chosen differently.** More parts means repacking the part number so the two flags move, moving the part table out of the shader's small set of uniform values and into a texture, and checking real phones. That is a change to the engine, not to a recipe. The kit has other ways to get many moving pieces: up to 48 tokens (game pieces moved and turned by their own offsets) and up to 96 levers in 6 groups (the piano keys).
>
> Where it lives: `part` in `src/kit.js`; the part number in the shader code of `src/effects.js`; `MAX_TOKENS` in `src/motion.js`.

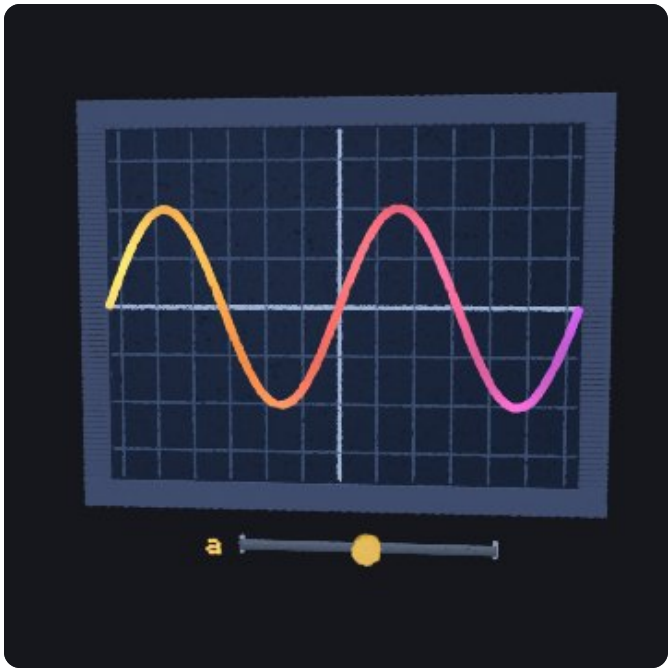
Could a toy use different parts?

EACH RECIPE'S CHOICE

The 15 is a limit on how many parts a toy can have. It comes from the 4-bit part number. Which pieces become parts is up to each recipe. A recipe makes a part with `k.part(name, { pivot, axis })` and puts shapes in it with `part:`. The windmill makes one part for all four sails. A recipe could instead make one part per sail, so each could turn on its own; it would then have used four of its 15.



The treasure chest: its lid is a part on a hinge.



The graph plotter: the a slider is a control.

Behaviors

Some motion is too fine for parts: a flame that flickers, snow that falls, grass that sways. A **behavior** moves each splat on its own on the graphics card: `kind:` `"flame"`, `"twinkle"`, `"sway"`, `"orbit"`, `"fall"`, `"wave"`, and more. **Morphs** give each splat a second place, `to: (c) => [x, y, z]`, and slide it there as one of four channels goes from 0 to 1. The Splat equation toy's motion is made of morphs.

A small, complete recipe

Here is a whole toy: a little windmill whose sails turn with a Wind slider, and a tap that blows a gust, spinning them once more around.

```
import { mix } from "../src/kit.js";

// A little windmill: a stone tower, a red cap, and four sails that turn in
// the wind. A tap blows a gust that spins them once more around.
export const RECIPES = {
  "little-windmill": {
    alive: true,
    options: [{ key: "sails", label: "Sail color", type: "color", default: "#f2eee4" }],
    controls: [
      { key: "wind", label: "Wind", type: "slider", default: 0.4 },
      { key: "gust", label: "Gust", type: "pulse", ease: 3 },
    ],
    action: { key: "gust", label: "Blow a gust" },
    drive(t, c, out) {
      // The wind turns the sails; a gust adds one extra turn, eased in and
      // out, so they end where they would have been anyway.
      const g = c.gust > 0 ? 1 - c.gust : 0;
      const extra = 2 * Math.PI * g * g * (3 - 2 * g);
      out.parts.sails = { angle: t * (0.3 + 2 * c.wind) + extra };
    },
    build(k, o) {
      // The tower: whitewashed stone, darker toward the ground.
      k.add(k.cone(0.5, 0.3, 1.6), {
        color: (c) => mix("#8b8f97", "#ece6d8", (c.p[1] + 0.8) / 1.6),
        interior: 0.1,
        core: "#6b5b4b",
      });
      // The cap.
      k.add(k.cone(0.36, 0.02, 0.4), { pos: [0, 1.0, 0], color: "#9c3b2c" });
      // The sails turn together, about an axis that points at you.
      const sails = k.part("sails", { pivot: [0, 0.6, 0.42], axis: [0, 0, 1] });
      k.add(k.sphere(0.07), { pos: [0, 0.6, 0.42], part: sails, color: "#4b3a2a" });
      for (let i = 0; i < 4; i++) {
        const a = (i * Math.PI) / 2;
        k.add(k.box(0.16, 0.9, 0.02), {
          pos: [-0.5 * Math.sin(a), 0.6 + 0.5 * Math.cos(a), 0.44],
          rot: [0, 0, (a * 180) / Math.PI],
          part: sails,
          color: o.sails,
        });
      }
    },
  },
};
```



This recipe, built by the kit.

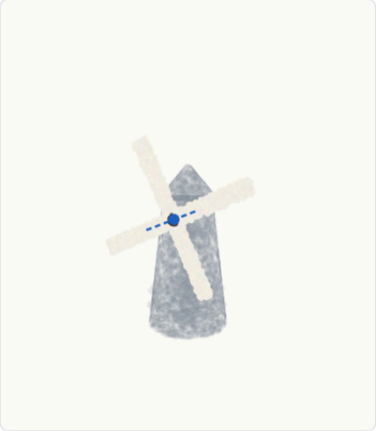


The Windmill on the shelf, a bigger toy built the same way.

Read it from the bottom up. `build` adds a cone for the tower (its color a function of height), a smaller cone for the cap, and a part called "sails" with a hub and four boxes, each turned a quarter turn more than the last. `drive` turns the sails part: steadily with the wind, plus one extra turn while the gust pulse runs, eased with the **smoothstep** curve $3g^2 - 2g^3$ (an S-shaped ramp that rises from 0 to 1 and is flat at both ends [7]) so it starts and ends gently. A full extra turn lands the sails where they would have been anyway, so nothing jumps when the gust ends.

Try it: which splats each line of the recipe made

This is the windmill from the recipe above, built right now by the real kit (12,000 splats). Point at a line of the recipe, or tap it, and the splats it made light up on the windmill; the rest fade. The line that makes the `sails` part lights up everything in that part. Then move the slider: the whole part turns as one solid piece about its pivot, and the tower and cap stay still.



```
k.add(k.cone(0.5, 0.3, 1.6), {
  color: (c) => mix("#8b8f97", "#ece6d8", (c.p[
  interior: 0.1,
  core: "#6b5b4b",
});
k.add(k.cone(0.36, 0.02, 0.4), { pos: [0, 1.0,
const sails = k.part("sails", { pivot: [0, 0.6,
k.add(k.sphere(0.07), { pos: [0, 0.6, 0.42], pa
for (let i = 0; i < 4; i++) {
  const a = (i * Math.PI) / 2;
  k.add(k.box(0.16, 0.9, 0.02), {
    pos: [-0.5 * Math.sin(a), 0.6 + 0.5 * Math.
    rot: [0, 0, (a * 180) / Math.PI],
    part: sails,
    color: o.sails,
  });
}
```

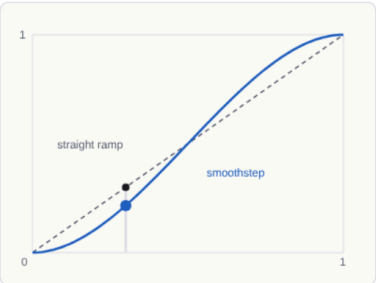
Sails turn 25°

```
everything in the sails part: the hub and
the four sails: 2,202 of 12,001 splats
(18%).
The part turns as one solid piece about
the dashed line (axis [0, 0, 1] through
the pivot). Move the slider.
```

Interactive in the web version: ryanjosephkamp.github.io/splashery/manual/#demo-windmill

Try it: smoothstep beside a straight ramp, and a turn that uses it

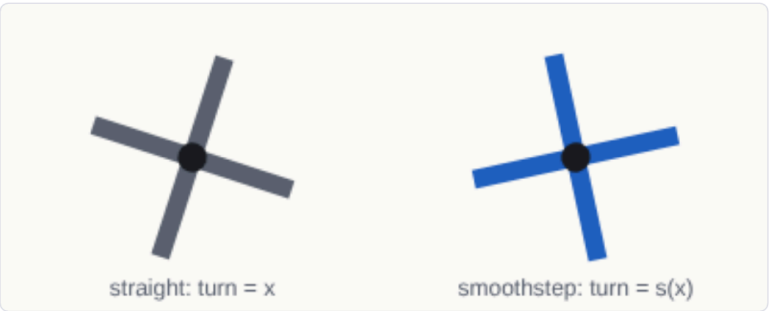
The curve $3x^2 - 2x^3$ on 0 to 1 is the solid line; a straight ramp is the dashed one. Drag the point, use the arrow keys on the curve, or use the slider. The two shapes below make the same one turn, one eased the straight way and one with smoothstep, so you can see the gentle start and stop on the shape itself.



x 0.30

Play the turn

```
x = 0.300
straight ramp:  0.3000
smoothstep:     3x² − 2x³ = 3 × 0.0900 −
2 × 0.0270 = 0.2160
extra turn:     108.0° (straight) and
77.8° (smoothstep)
```



Interactive in the web version: ryanjosephkamp.github.io/splashery/manual/#demo-smoothstep

Run it yourself

You can run the windmill on your own computer in five steps. You need a copy of the Splashery source (download it from GitHub) and Python 3.

1. **Save the recipe as a pack.** Copy the module above into `src/packs/little-windmill.js`, or copy manual/example-recipe.js. Change its first line to `import { mix } from "../kit.js";`. A pack lives in `src/packs/`, so the kit is one folder up. (The copy in `manual/` says `../src/kit.js` because it sits beside `src/`, not in it.)
2. **Add its catalog entry.** In `src/toys.js`, find the line `export const TOYS = [` and paste this right after it:

   ```
   {
     id: "little-windmill",
     label: "Little windmill",
     category: "objects",
     kind: "kit",
     pack: "little-windmill",   // loads src/packs/little-windmill.js
   },
   ```

   The `id` must match the key in the pack's `RECIPES`.
3. **Serve the folder.** In the source folder, run `python3 -m http.server 8000 --bind 127.0.0.1`. The page needs a server because browsers will not load ES modules from a file address.
4. **Open the toy.** Go to `http://127.0.0.1:8000/embed/index.html?toy=little-windmill`. The embed page opens a catalog toy by its id. The windmill turns by itself.
5. **Tap the windmill.** The tap runs the recipe's action, the **Blow a gust** pulse: the sails spin one extra turn, easing in and out, and end where they would have been.

The same steps, as a prompt. If you work with a coding assistant, paste this into it, from inside your copy of the source folder:

```
In this copy of the Splashery source, please do exactly these steps and nothing else:
1. Create src/packs/little-windmill.js as a copy of manual/example-recipe.js, but change its
   first line to: import { mix } from "../kit.js";
2. In src/toys.js, right after the line "export const TOYS = [", add this entry:
   { id: "little-windmill", label: "Little windmill", category: "objects", kind: "kit",
     pack: "little-windmill" },
3. Start a local server in the source folder: python3 -m http.server 8000 --bind 127.0.0.1
4. Tell me to open http://127.0.0.1:8000/embed/index.html?toy=little-windmill and what to
   expect: the sails turn by themselves, and a tap spins them one extra turn.
```

> The Splat equation toy's text box takes equations (Level 3), not JavaScript recipes. A recipe has to be a module in `src/packs/`, as above.

CHAPTER 3

The splat equation language

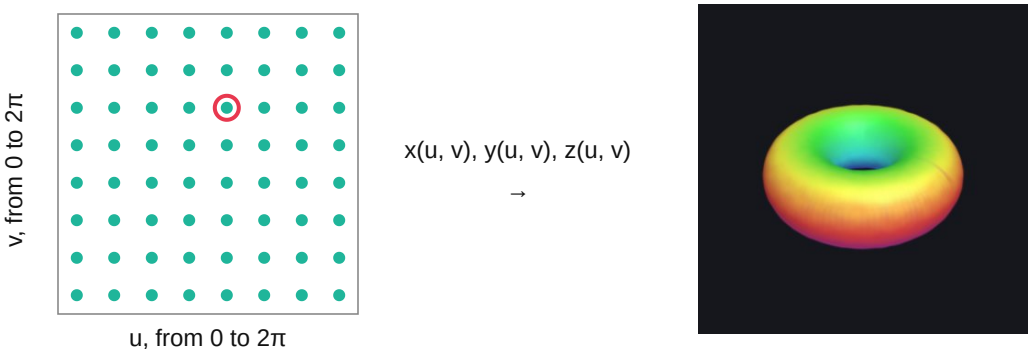
Recipes are powerful, but they are code, and code from a stranger can do things you did not ask for. The **Splat equation** toy on the math shelf gives you a smaller language that is only math. You write where every splat goes and what color it is, as equations, and nothing else can happen. That is why a link with your equations in it is safe to share.

Open it with any link in the gallery (Level 4). Type in the box under **Your own splat program** in the Toy tab and press **Splat it**. The Program list has eight famous shapes to start from.

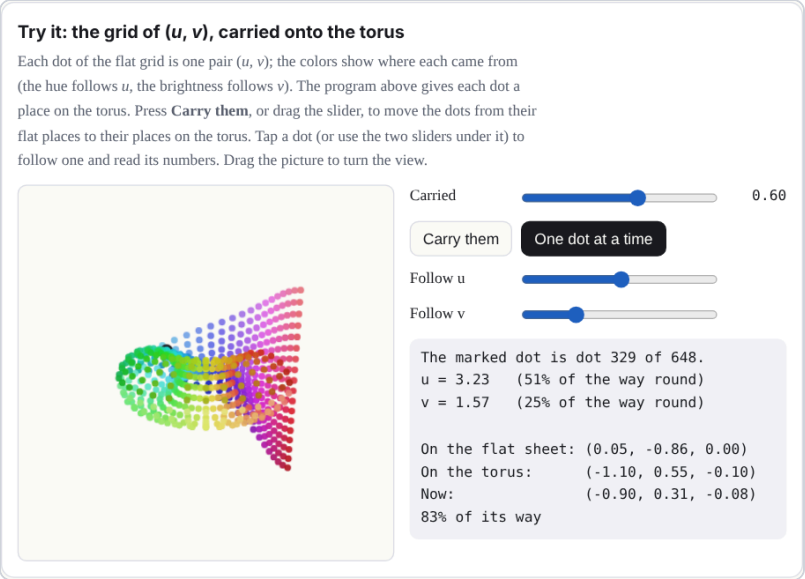
The idea: every splat gets its own u and v

The toy lays out its splats on a grid of two numbers, *u* and *v*, each running over a range you choose. Think of them as the coordinates of a flat sheet of points, laid out before the equations bend it into shape. They are the parameters of a parametric surface [9], not a substitution: nothing is replaced by *u* or *v*; each splat simply has its own pair. Your equations turn each splat's (*u*, *v*) into a place in space, (*x*, *y*, *z*). If *u* goes around a circle and *v* around a smaller circle, the grid wraps into a torus:

```
x = (2 + cos v)·cos u
y = sin v
z = (2 + cos v)·sin u
u = 0 .. 2pi
v = 0 .. 2pi
```



Each dot on the grid is one splat with its own (*u*, *v*). The equations carry every dot to its place on the torus.



Interactive in the web version: ryanjosephkamp.github.io/splashery/manual/#demo-uvtorus

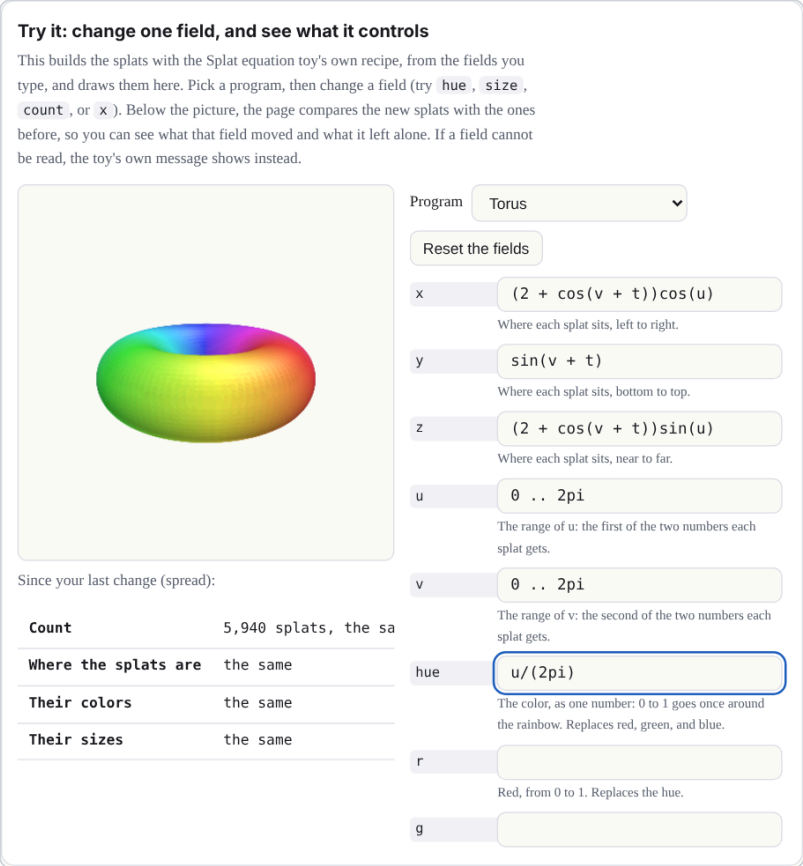
The fields

A program is a list of **fields**, each written `name = value`. You can type one field or several, split by `;` (a text file can put one on each line). You do not have to retype everything. A field you leave out keeps what is already showing: if you type only a new `hue`, the shape stays and only the color changes.

Show the table: the fields of a program

Field	What it is	If left out
`x`, `y`, `z`	Where each splat goes, as equations in *u*, *v*, and *t*. *y* points up.	Needed
`u`	The range of *u*, like `0 .. 2pi`.	Needed
`v`	The range of *v*.	`0 .. 1`
`hue`	Color as a place on the color wheel: 0 is red, 1/6 yellow, 1/3 green, 1/2 cyan, 2/3 blue, 5/6 magenta, and 1 is red again. Any number works; it wraps around every whole turn, so −0.1 gives the same hue as 0.9.	Use *r*, *g*, *b*
`r`, `g`, `b`	Color as red, green, and blue, each from 0 to 1. Setting any of them turns *hue* off; setting *hue* turns them off.	0.5 each (gray)

Field	What it is	If left out
`size`	How big each splat is, in the same units as *x*, *y*, and *z*: about the σ of its bell. It may use *u*, *v*, and *t*.	`0.05`
`count`	How many splats: 100 to 10,000.	`4000`
`spread`	`grid` spaces the splats evenly; `random` scatters them at random over the ranges (a galaxy, a firework).	`grid`



Interactive in the web version: ryanjosephkamp.github.io/splashery/manual/#demo-fields

Why this number? Twelve fields

OUR CHOICE, NEAR A LIMIT The toy chose which fields; saved links limit how many a toy can keep.

The toy's program is saved in its link as options, and a link keeps at most 16 option keys per toy. The twelve fields, plus the program picker, the splat style (Solid, Fine, or Dots), and the Light and shade switch, come to 15. One more key would fit; two would not.

If it were chosen differently. More fields means more option keys, so either the limit of 16 rises (a change to how every saved link is read, which must keep old links loading) or fields share a key.

Where it lives: `normalizeOptions` in `src/state.js`; `FIELDS` in `src/packs/splat-equation.js`.

Could the equation toy have different fields?

NOT TESTED YET Nothing below is built.

Yes. Each field you add costs one of the 16 option keys a saved link keeps, and the toy uses 15 today. Candidates:

- **Opacity**, so a splat could be see-through.
- **A stretch and a turn for each splat**, which would make an oriented, flattened splat instead of a round one.
- **Glow**, a brightness that does not depend on the lighting.
- **A part for each point**, so each point could move on its own. This is the costly one: toys are limited to 15 parts.

Flexibility

The reader is forgiving about how you write a program:

- **Alternative names.** `color` for `hue`; `red`, `green`, and `blue` for `r`, `g`, and `b`; `splats` for `count`; `x(u,v,t)` for `x` (and likewise `y(u,v,t)` and `z(u,v,t)`); also `colour` and `n`, which the code accepts.
- **Three ways to write a range.** `0 .. 2pi`, `0 to 2pi`, or `0, 2pi`. (Three dots, `0 ... 2pi`, work like two.)
- **Capitals do not matter.** Names are not case-sensitive: `Hue` is `hue`.
- **Notes.** Anything after a `#` on a line is a note, and the reader skips it.

Expressions

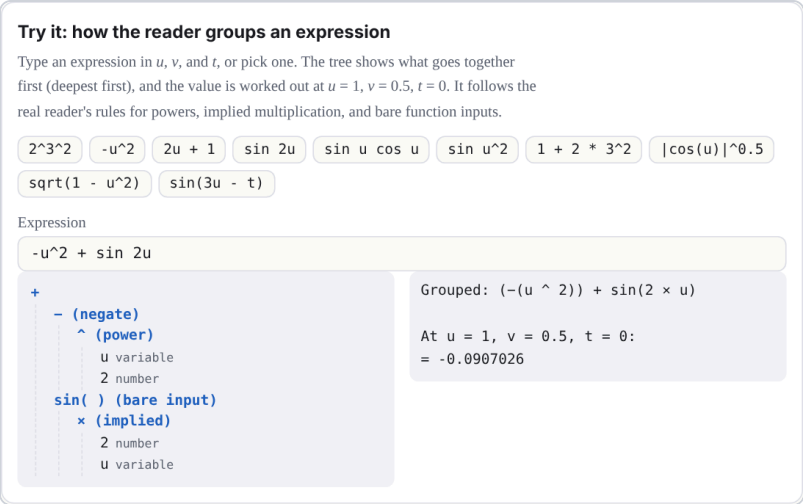
The equations are read by the same small, safe reader as the Graph and Surface plotters. It understands:

- **Numbers**: `2`, `0.5`, `.25`.
- **Variables**: *u*, *v*, and *t*. Range ends and the count are numbers only.
- **Constants**: `pi` (or π) and `e`.
- **Operators**: `+ − × ÷ ^`, written `+ - * / ^` (and `**`, `·`, `²`, `³`), with brackets `( )`, `[ ]`, and `{ }` (the reader treats all three kinds alike), and `|u|`

for absolute value.

- **Functions**: `sin cos tan asin acos atan sinh cosh tanh exp log` (natural; `ln` too), `log10 sqrt` (or √), `abs floor`, and `min` and `max` of two or more values. The inverse trig functions also go by `arcsin`, `arccos`, and `arctan`.
- **Shortcuts**: `2u` means 2 × *u*, and `3sin(u)` and `(u + 1)(v − 1)` multiply. A function can also skip its brackets when its input is a short product: a run of numbers and names, each with an optional power. The input stops at the next function, at a bracket, or at a plus or minus. So `sin 2u` is sin(2*u*), `sin u cos u` is sin(*u*) × cos(*u*), and `sin u^2` is sin(u^2), not (sin *u*)². A minus sign cannot start a bare input (write `sin(-u)`), and `min` and `max` always need brackets.

Powers go first, then × and ÷, then + and −, and a power binds to the right: `2^3^2` is 2^{9}, and `-u^2` is $-(u^2)$. Each field can be up to 240 characters long.



Interactive in the web version: ryanjosephkamp.github.io/splashery/manual/#demo-reader

Why this number? The 240-character limit

OUR CHOICE Nothing in the math or the browser needs it.

A program lives in the toy's link, and a link holds up to 12 fields, so a limit keeps links short enough to paste into a message. The longest field of the toy's own programs is 53 characters. The limit was 120 until October 10, 2026, when it was raised to 240, for people who write longer formulas. A link written with the old limit loads the same as before.

If it were chosen differently. The link could carry much more (a text option may be up to 24,000 characters), so the limit is not what stops longer fields. Two other guards would be next: expressions nest at most 32 deep and have at most 400 pieces. The cost of longer fields is longer links and slower builds, because each expression is worked out for every point at 13 moments. At 240 characters a field can take up to

twice as long to build as at 120, and the longest possible link grows from about 1,400 to about 1,900 characters once compressed (`docs/lab/EQUATION-FIELDS.md`). The graph and surface plotters keep 120.

Where it lives: `FIELD_MAX` in `src/packs/splat-equation.js` (240); `MAX_LENGTH` (120, still the plotters' limit), `MAX_DEPTH` , and `MAX_NODES` in `src/equation.js` ; `TEXT_MAX` in `src/state.js` .

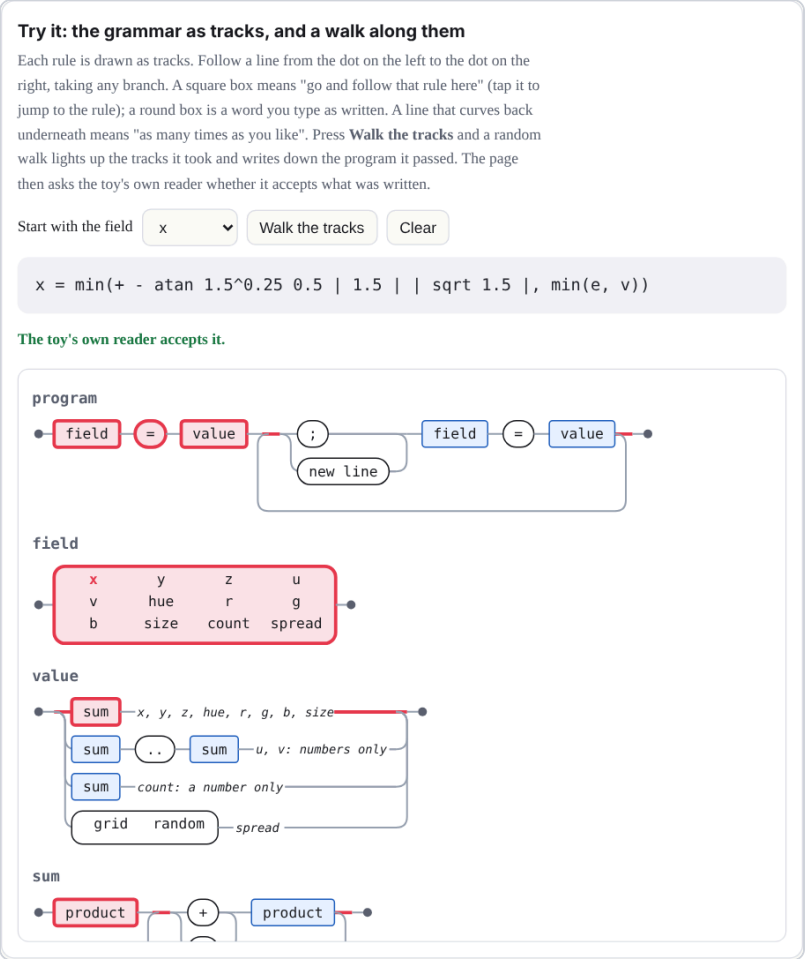
The grammar

Here is the core grammar. Note: words in quotes must be typed exactly as written below.

A grammar is a short list of rules for what counts as a valid program. Its notation follows EBNF, a standard way to write such rules [8]:

- `=` means "is made of": the name on the left is built from the pattern on the right.
- `|` means "or". (`"|"` in quotes is the absolute-value sign you type.)
- `{ }` means "repeat what is inside any number of times, including none".
- `[ ]` means "optional": what is inside may be there or not.
- Quotes mark text you type exactly. Plain parentheses group choices together. Words with no quotes, such as `sum` , are names of other rules. A note in parentheses at the end of a line is only a comment.
- An **atom** is the smallest piece of an expression that can stand on its own: a number, a name, a function with its input, or something in brackets.

```
program    = field "=" value { ( ";" | new line ) field "=" value }
field      = "x" | "y" | "z" | "u" | "v" | "hue" | "r" | "g" | "b"
           | "size" | "count" | "spread"
value      = sum                          (x, y, z, hue, r, g, b, size)
           | sum ".." sum                 (u, v: numbers only, and no min or max)
           | sum                          (count: a number only)
           | "grid" | "random"            (spread)
sum        = product { ( "+" | "-" ) product }
product    = signed { ( "*" | "/" ) signed | power }   (nothing between: 2u)
signed     = { "-" | "+" } power
power      = atom [ "^" signed ]
atom       = number | "u" | "v" | "t" | "pi" | "e"
           | function "(" sum ")"
           | function bare                (sin 2u, sin u cos u, sin u^2)
           | ( "min" | "max" ) "(" sum "," sum { "," sum } ")"
           | bracket | "|" sum "|"        (not directly inside another | |)
bracket    = "(" sum ")" | "[" sum "]" | "{" sum "}"
bare       = simple { simple }            (stops at the next function or bracket)
simple     = ( number | "u" | "v" | "t" | "pi" | "e" ) [ "^" signed ]
function   = "sin" | "cos" | "tan" | "asin" | "acos" | "atan"
           | "arcsin" | "arccos" | "arctan" | "sinh" | "cosh" | "tanh"
           | "exp" | "log" | "ln" | "log10" | "sqrt" | "abs" | "floor"
                                          (one input; min and max are above)
```



Interactive in the web version: ryanjosephkamp.github.io/splashery/manual/#demo-railroad

Time

t is time. A tap on the toy plays it from 0 to 2π in about 4 seconds, and the **t** slider in the Toy tab holds it anywhere in between. Anything with *t* in it moves: `u + t` turns a shape, `sin(t)` breathes, and `sin(3u − t)` sends a wave around.

Why this number? The two clocks

OUR CHOICE Both of them.

There are two clocks. The math clock, *t*, runs from 0 to 2π, one full circle, so that `sin` and `cos` of it come back exactly to where they started, and loops are seamless. The wall clock is real time: a play of *t* lasts about 4 seconds, whatever 0 to 2π means to the math. The speed is eased, so a play starts and ends still.

If it were chosen differently. A different range for *t* would change what a loop looks like (a range of 1 would make `sin(t)` stop mid-wave). A different play time would only change how fast it looks.

Where it lives: `PLAY_SECS` in `src/packs/splat-equation.js` .

Behind the scenes, the toy builds your shape at 12 moments, t = 0, π/6, 2π/6, and so on, and each copy slides every splat in a straight line to its place in the next copy (a morph, Level 2), so every splat is at its computed place at those 12 moments (up to the packing error described below) and slides in a straight line in between. A shape that turns dips very slightly inward between moments, about 3 percent. (With labs on, a **Moments of t** choice sets 13, 14, or 15 moments instead; 12 stays the default, so older links build as before.)

Why this number? Twelve moments

OUR CHOICE, NEAR A LIMIT Our choice; each moment also costs one of the 15 parts.

Twelve copies of 10,000 points (the most `count` allows) make 120,000 splats, the Low tier's largest budget, so even the weakest device can hold the biggest program. (The kit keeps 2 percent in reserve, so on that device each copy is cut to 9,800 points.) Each moment is its own part (Level 2), and 12 parts plus the body fit in the limit of 15.

If it were chosen differently. More moments make a turning shape rounder, because a splat slides in a straight line between moments and so dips inward in between. The worst dip is 1 − cos(π ÷ moments): 3.4 percent at 12 moments, 2.2 percent at 15, and 0.9 percent at 24. But the points per copy shrink on the weakest device: 9,800 at 12, 7,840 at 15, and 4,900 at 24. And because each moment is a part, 15 is the most without a bigger change. Color and size step at the moments rather than sliding, so more moments would smooth color changes too. The moments demo below lets you try every count. The labs choice **Moments of t** now offers 12, 13, 14, or 15; the Kit lab measured the worst dip of the splats at 3.41, 2.91, 2.51, and 2.19 percent (`docs/lab/MOMENTS.md`).

Where it lives: `KNOTS` (the default, 12) and `MOMENTS` (the choices) in `src/packs/splat-equation.js` ; the points per copy are floor(0.98 × budget ÷ moments).

Why is each moment a part, and what else could work?

NOT TESTED YET The alternatives below are ideas, not built.

The toy builds 12 complete copies of the shape, one for each moment of t. It shows one copy at a time and slides each into the next with the morph channel, so each copy needs its own part number, to be shown or hidden on its own. Other ways to do it:

- **Store the moments in a keyframe texture** and blend them in the shader. This needs no parts and allows more moments, but it is a change to the engine.
- **Compile the formulas to a shader**, so *t* is continuous and there are no copies at all. This works because every splat is a pure function of its own (*u*, *v*, *t*), with nothing shared between splats, so the graphics chip could work them all out at once. It is the larger experiment.
- **Recompute on the processor every frame.** This is too slow at 10,000 or more points, with a fresh sort every frame.

Why this number? Twelve bits

FORMAT LIMIT It follows from how much room each splat has.

A moment's target place for each splat is stored as three whole numbers, one per axis, each of 12 bits: 4,096 steps across a range of ±2 toy units. These are real binary digits, not a figure of speech. A toy is fitted to a sphere 1.9 units across, so any move fits, and one step is about 1/1,900 of the toy's width. Rounding to a step is the "tiny error" the manual mentions: at most half a step.

Each splat's animation data is four 32-bit numbers, and two of them are already taken (the part number with its flags, and the kind of motion). That leaves two. A 32-bit float holds any whole number up to 2^{24} exactly, and two 12-bit numbers make 24 bits, so one float carries two axes. The third axis and a channel number share the other.

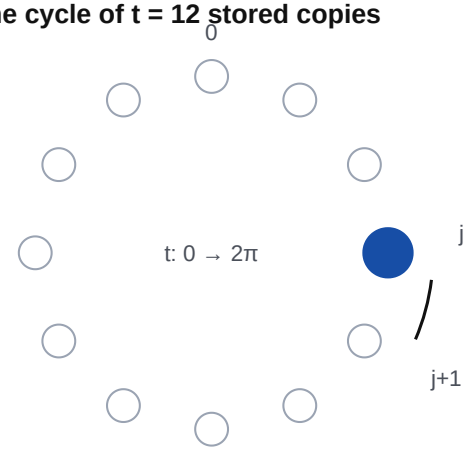
```
z = qx × 4096 + qy          (two 12-bit numbers, exact)
w = qz + 4096 × channel     (one 12-bit number and the channel)
```

If it were chosen differently. More bits would not fit two axes in one float, so the toy would need another float per splat (every float is already spoken for) or a texture. Fewer bits would make the steps visibly coarse.

Where it lives: `encodeMorphs` in `src/kit.js` ; `MORPH_RANGE` in `src/effects.js` .

Approximations in time. The shape is sampled at 12 moments, not solved for every *t*. Only each splat's *position* slides toward the next copy, and that target is packed into 12 bits per axis over a fixed range, so tiny errors are possible. Its color, size, and rotation are built separately for each copy, so they can step at a copy boundary instead of blending. The "count" you type is the number of points sampled for each copy, not the total number of splats in the toy (Figure 3).

One cycle of t = 12 stored copies



Twelve copies, built at t = 0, 2π/12, 4π/12, …

Copy j shows and slides toward copy j + 1.

The other eleven are stored but hidden.

Where the splats go (the detail budget)

toy budget by device tier: Low 120,000 · Mid 240,000 · High 300,000 · Max 400,000

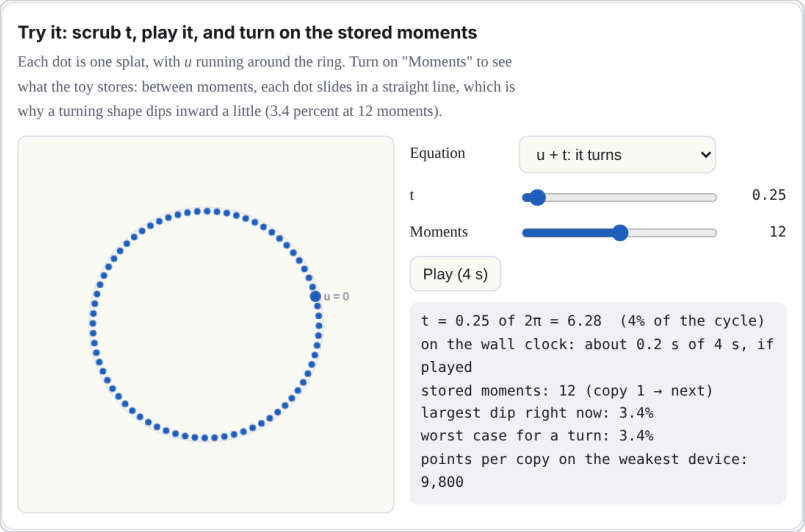
per copy: ⌊0.98 × toy budget ÷ 12⌋ (Low 9,800 … Max 32,666)

points per copy: the smaller of count and that budget

Fine: up to 4 × count, still within the budget

grid rounds down; points with no value are dropped

Figure 3. A program with *t* is stored as 12 copies, so the number of splats in memory is about 12 times the points in one copy. The device's tier sets the budget at the top; your count, the Fine setting, and the grid can only lower the points per copy.



Interactive in the web version: ryanjosephkamp.github.io/splashery/manual/#demo-moments

> **Make it loop.** Because *t* ends at 2π, a shape made of `sin(t)`, `cos(t)`, and angles like `u + t` ends exactly where it began. A shape with plain `t` in it (like `y = t`) jumps back at the end. `sin(t/2)` goes out and back once; `sin(2t)` twice.

A program without *t* stands still. Then a tap clears it and draws the splats again one by one, in the order they are made: the first value of *u* with every *v*, then the next *u*, and so on.

What the toy does for you

- **It fits the view.** The shape at its sampled moments of *t* is scaled to fit, so your units don't matter; only the sizes compared with the shape do.
- **It spaces the grid.** With `spread = grid`, the side of the shape that is longer gets more splats, so they sit about evenly apart. A shape that uses only *u* is a curve, and all the splats go along it.
- **It lights surfaces.** For a grid surface in Solid or Fine mode, each splat is a flat disc facing out from it (along the cross product of its slopes in *u* and *v*), lit from above. The **Light and shade** switch turns the lighting off, so you see your colors exactly. This is a simple shade, not physical lighting: your base color is multiplied by 0.62 + 0.48 × (how squarely the disc faces a fixed light) and a small highlight is added.
- **It skips impossible points.** Where an equation has no value (the square root of a negative number, a division by zero, the log of 0), that splat is left out. So is a point more than three times farther out than nearly all the others. A hue that can't be worked out becomes gray, a red, green, or blue that can't becomes 0, and a size that

can't becomes a small default. If a splat's next moment is impossible, it holds still instead of flying off.

- **It explains mistakes.** Each field that can't be read gets one plain message, like *“z: That can't be drawn: I don't know “q”.”*, and the shape on the stage stays as it was. Others are *“the number “1.2.” has two points.”* and *“w” is not a field. Use x, y, z, u, v, hue, r, g, b, size, count, spread.*

Why it is safe

The reader never runs what you type. It breaks the text into numbers, the letters *u*, *v*, and *t*, the names in the lists above and the signs, and builds a small tree of arithmetic from them. There is no way to name anything else, so the worst a program can do is draw a strange shape. Your program lives in the toy's link ("Copy link" in the Share tab), so you can send it to anyone.

CHAPTER 4

A gallery of programs

The first eight are the toy's own programs; the rest are small ideas to copy and change. Each button opens the toy with that program; then tap it to play *t*. While the toy is being tried out, it is on the shelf only when Splashery is opened with `?labs=1`, but these links always open it.

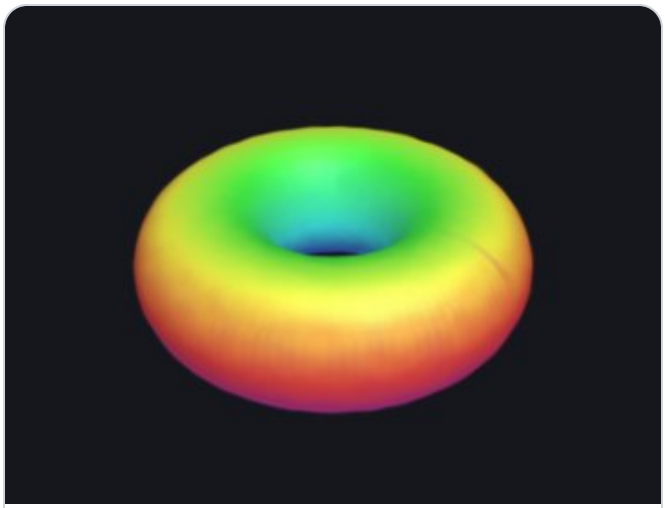


Sphere

A sphere from two angles, u around and v from pole to pole, spinning with u + t. Six stripes from floor(6u/(2pi)).

```
x = sin(v)cos(u + t)
y = cos(v)
z = sin(v)sin(u + t)
u = 0 .. 2pi
v = 0 .. pi
hue = floor(6u/(2pi))/6 + 0.04
size = 0.035
count = 7000
```

Open it in the toy



Torus

A tube around a ring. v + t rolls the tube around itself, so it turns inside out as you watch.

```
x = (2 + cos(v + t))cos(u)
y = sin(v + t)
z = (2 + cos(v + t))sin(u)
u = 0 .. 2pi
v = 0 .. 2pi
hue = v/(2pi)
size = 0.09
count = 6000
```

Open it in the toy

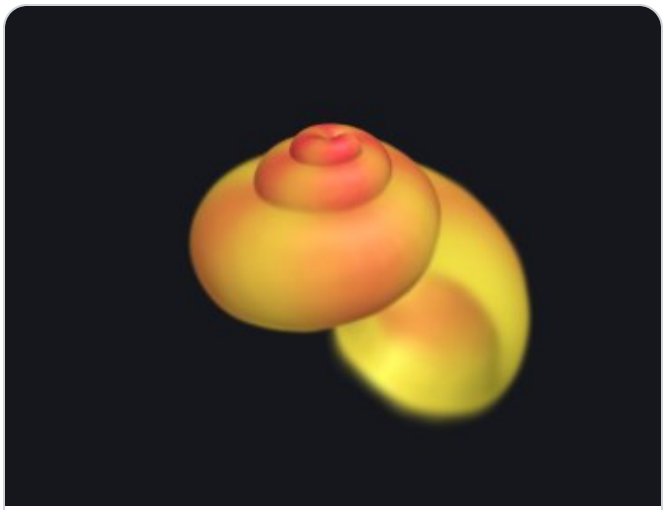


Möbius strip

A band with a half twist: cos(u/2) turns the band over once as u goes around. The colors flow along it.

```
x = (1 + v cos((u + t)/2))cos(u + t)
y = v sin((u + t)/2)
z = (1 + v cos((u + t)/2))sin(u + t)
u = 0 .. 2pi
v = -0.4 .. 0.4
hue = u/(2pi)
size = 0.035
count = 4000
```

Open it in the toy



Seashell

A tube that grows by exp(0.1u) as it winds three times around, turning with u + t. Its splats grow too.

```
x = exp(0.1u)cos(u + t)(1.2 + cos v)
y = exp(0.1u)(sin(v) - 2.4) + 3
z = exp(0.1u)sin(u + t)(1.2 + cos v)
u = 0 .. 6pi
v = 0 .. 2pi
hue = 0.02 + 0.1u/(6pi) +
0.03cos(4u)
size = 0.14exp(0.1u)
count = 10000
```

Open it in the toy

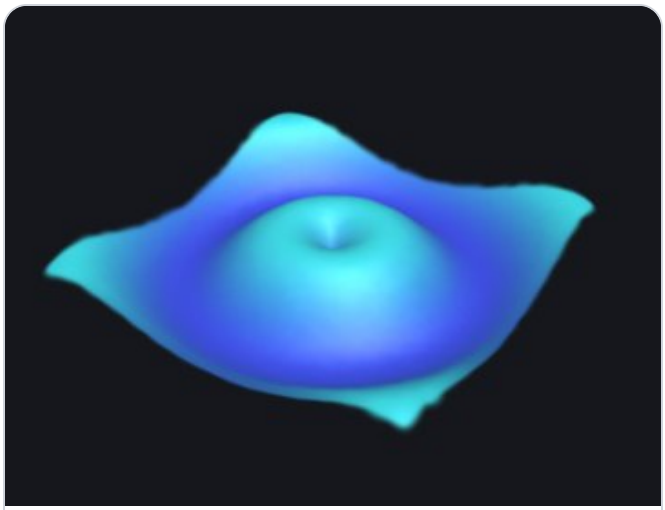


Trefoil knot

The simplest real knot, a curve in u alone, turned about the vertical by the rotation formulas in t.

```
x = (sin(u) + 2sin(2u))cos(t) -
sin(3u)sin(t)
y = cos(u) - 2cos(2u)
z = -(sin(u) + 2sin(2u))sin(t) -
sin(3u)cos(t)
u = 0 .. 2pi
v = 0 .. 1
hue = u/(2pi)
size = 0.16
count = 2500
```

Open it in the toy

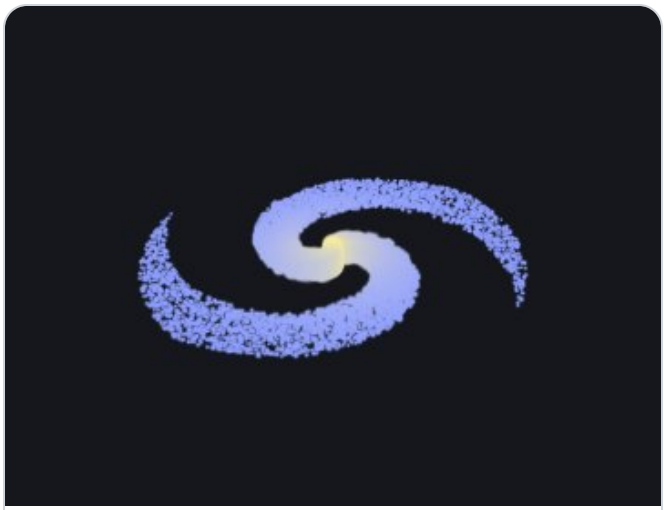


Wave

A flat sheet in u and v with ripples running out from the middle: sin(distance − t).

```
x = u
y = 0.45sin(2sqrt(u^2 + v^2) - t)
z = v
u = -3 .. 3
v = -3 .. 3
hue = 0.58 - 0.07sin(2sqrt(u^2 +
v^2) - t)
size = 0.07
count = 6000
```

Open it in the toy

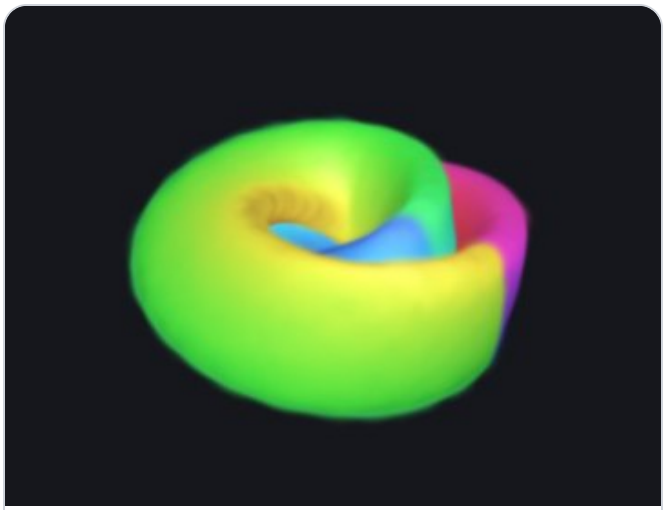


Spiral galaxy

Two arms from floor(v) (0 or 1, half a turn apart), scattered at random, turning once. Colors from r, g, and b.

```
x = u^1.5 cos(pi floor(v) + 5u +
1.2(v - floor(v)) - t)
y = 0.06(1 - u)sin(91v + 37u)
z = u^1.5 sin(pi floor(v) + 5u +
1.2(v - floor(v)) - t)
u = 0.02 .. 1
v = 0 .. 2
r = 0.55 + 0.45(1 - u)^2
g = 0.62 + 0.33(1 - u)^2
b = 1 - 0.45(1 - u)^3
size = 0.006 + 0.012(1 - u)^2
count = 10000
spread = random
```

Open it in the toy



Klein bottle (figure eight)

The figure-eight Klein bottle: a figure-eight tube around a ring, with a half twist, so it has no inside or outside.

```
x = (2 + cos(u/2)sin(v + t) -
sin(u/2)sin(2v + 2t))cos(u)
y = sin(u/2)sin(v + t) +
cos(u/2)sin(2v + 2t)
z = (2 + cos(u/2)sin(v + t) -
sin(u/2)sin(2v + 2t))sin(u)
u = 0 .. 2pi
v = 0 .. 2pi
hue = v/(2pi)
size = 0.13
count = 8000
```

Open it in the toy



Hello, circle

The smallest program: one loop of u makes a ring of 200 splats. No v, no t.

```
x = cos(u)
y = sin(u)
z = 0
u = 0 .. 2pi
hue = u/(2pi)
size = 0.06
count = 200
```

Open it in the toy

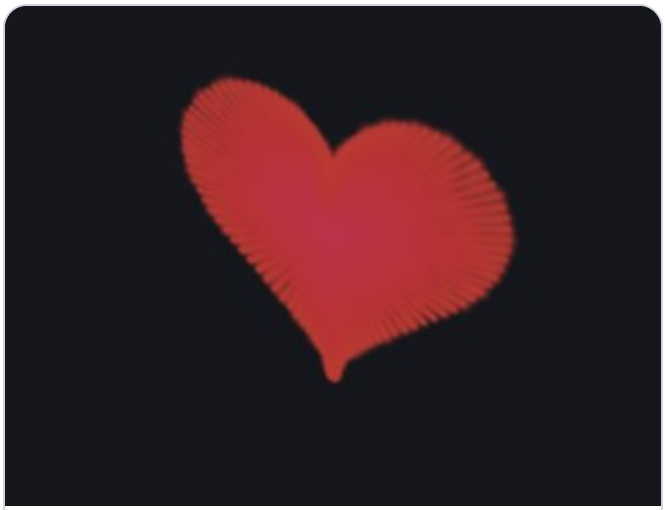


Bouncing spring

Three turns of a helix whose height breathes with sin(t).

```
x = cos(u)
y = (0.09 + 0.035sin(t))u - 0.9
z = sin(u)
u = 0 .. 6pi
hue = 0.08 + u/(6pi)/3
size = 0.07
count = 1500
```

Open it in the toy



Beating heart

A famous heart curve, filled in by v from the middle out, beating twice per cycle.

```
x = v(1 + 0.08sin(2t))16sin(u)^3
y = v(1 + 0.08sin(2t))(13cos(u) -
5cos(2u) - 2cos(3u) - cos(4u))
z = 0
u = 0 .. 2pi
v = 0 .. 1
hue = 0.97 + 0.04v
size = 0.35
count = 4000
```

Open it in the toy

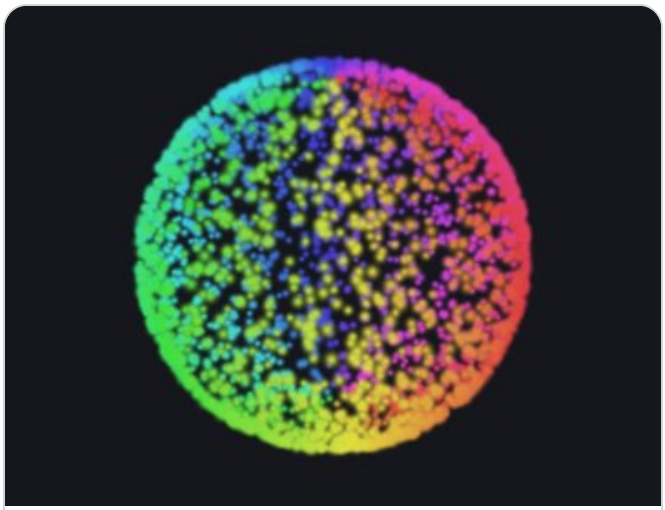


Rainbow ribbon

Color from r, g, and b: three cosines about a third of a turn apart make a rainbow along u.

```
x = u
y = 0.4sin(2u - t)
z = v
u = -3 .. 3
v = -0.4 .. 0.4
r = 0.5 + 0.5cos(u)
g = 0.5 + 0.5cos(u - 2)
b = 0.5 + 0.5cos(u + 2)
size = 0.05
count = 4000
```

Open it in the toy



Firework

Splats scattered at random over a sphere (v is the height, from −1 to 1) fly out, droop, and come back.

```
x = (0.2 + sin(t/2))sqrt(1 -
v^2)cos(u)
y = (0.2 + sin(t/2))v -
0.35sin(t/2)^2
z = (0.2 + sin(t/2))sqrt(1 -
v^2)sin(u)
u = 0 .. 2pi
v = -1 .. 1
hue = u/(2pi)
size = 0.018
count = 3000
spread = random
```

Open it in the toy

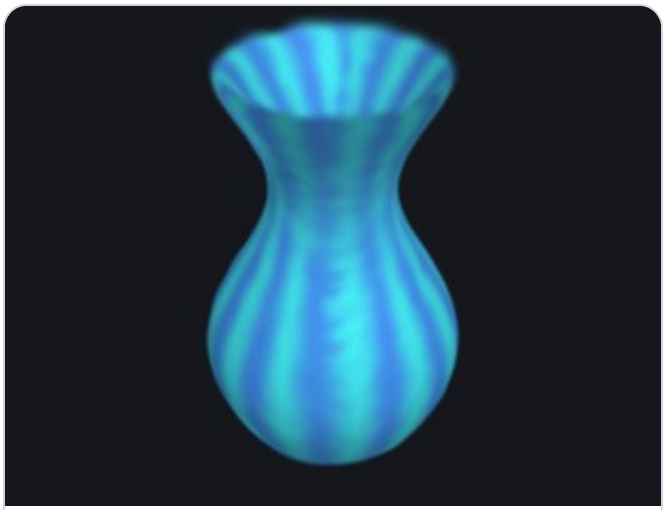


Opening flower

Five petals from cos(5u), and a cup that opens and closes with sin(t/2).

```
x = v(0.6 + 0.4cos(5u))cos(u)
y = (0.15 + 0.6sin(t/2))v^2
z = v(0.6 + 0.4cos(5u))sin(u)
u = 0 .. 2pi
v = 0 .. 1
hue = 0.92 - 0.12v
size = 0.035
count = 5000
```

Open it in the toy



Vase (no t)

A shape without t stands still, so a tap draws its splats again one by one, in the order they are made.

```
x = (0.5 + 0.2sin(3v))cos(u)
y = v
z = (0.5 + 0.2sin(3v))sin(u)
u = 0 .. 2pi
v = 0 .. 2.2
hue = 0.55 + 0.05sin(12u)
size = 0.045
count = 4000
```

Open it in the toy

Sixty more programs

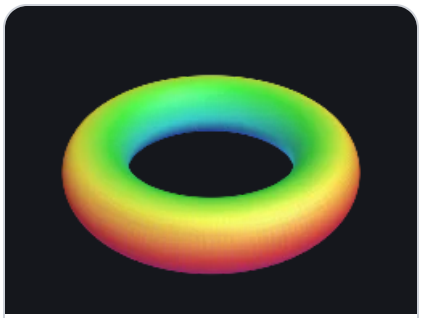
A second gallery, in six groups of ten, written and then checked by hand: each program reads, builds, and looks like its title. A few are honest look-alikes or samples of a named system, and say so; none is a full solution of the system it names. Each card has the picture, one line about the shape, the program (folded), and a **Run it** link that opens the Splat equation toy with that program, in a new tab. Every program fits the toy's own limits: 12 fields, 240 characters each (these programs are all under 120, so they would fit the old limit too), and at most 10,000 points.

Classic surfaces (10 programs)



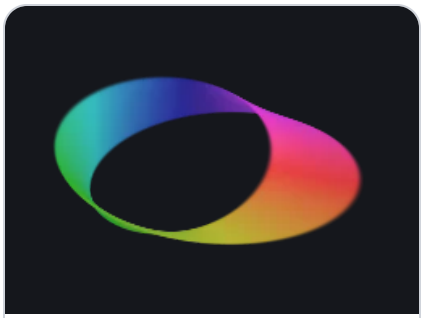
Spun sphere

A sphere turns once around its vertical axis while its latitude stripes stay in place.



Rainbow torus

A round tube wraps around a ring as its colors follow the small circle.



Möbius ribbon

A narrow band turns through one half twist before its edges meet.

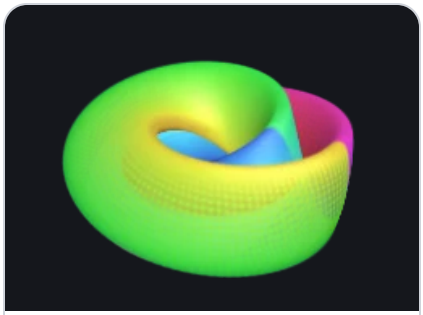
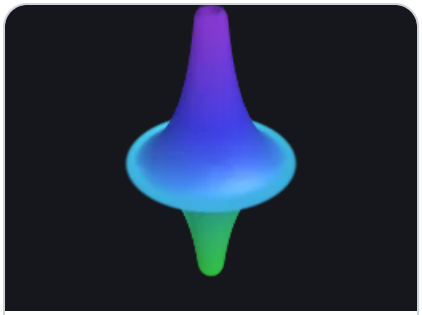


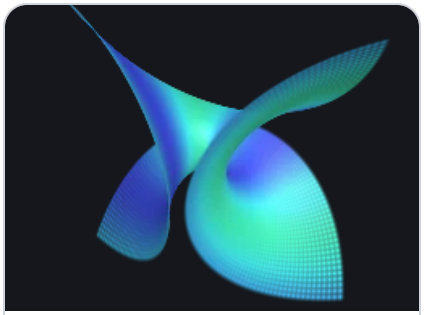
Figure-eight Klein bottle

This immersed Klein bottle crosses itself where its neck passes through the body.



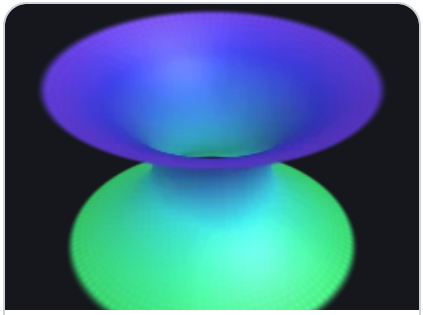
Pseudosphere

Two tractrix horns meet at a sharp rim: a surface that curves the opposite way from a sphere.



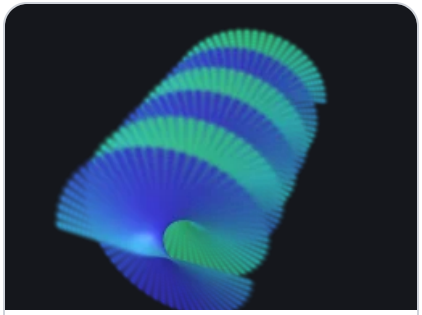
Enneper surface

A simple polynomial surface curls into three spreading lobes.



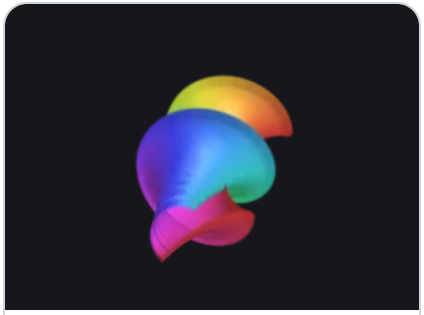
Catenoid

A catenoid pinches between two wide circular rims.



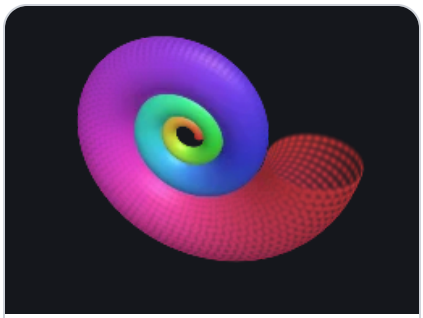
Helicoid

A ruled helicoid spirals upward like a ramp with no outer rim.



Dini surface

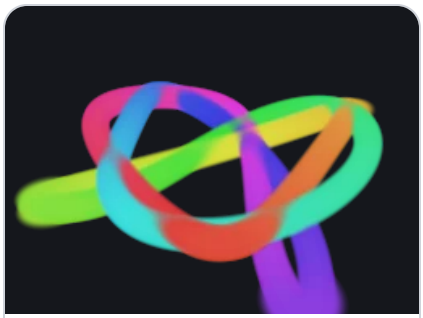
Dini’s surface twists a trumpet shape into a tight spiral.



Shell growth

A logarithmic tube grows wider as it winds away from its narrow tip.

Curves as tubes or beads (10 programs)



Lissajous knot

Three harmonics trace a closed, crossing space curve with a fine bead tube.



Trefoil tube

A tube follows the classic three-crossing trefoil knot and rotates in place.



Cinquefoil knot

Five over-and-under crossings wind around a torus in this thicker knot.



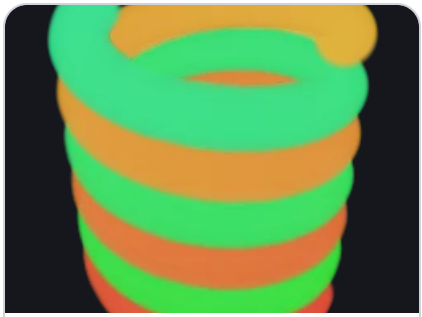
Rose curve beads

A six-petal planar rose becomes a dotted three-dimensional curve.



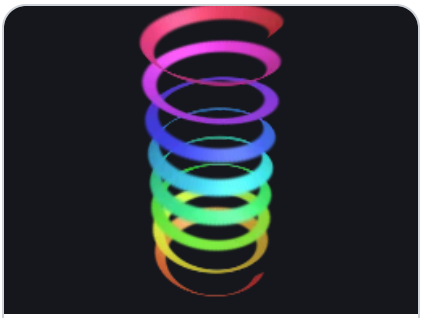
Figure-eight knot

This closed bead tube traces the simplest knot that is not a torus knot.



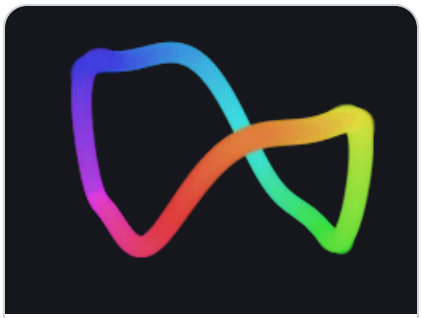
Double helix

Two spiral strands wind together around a long axis.



Slinky coil

A spring winds upward in eight turns with a narrow tube around its wire.



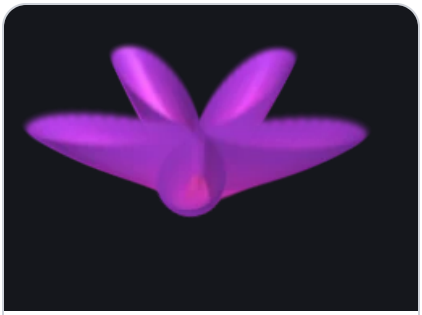
Lorenz-like look-alike

This butterfly loop is a Lorenz-like look-alike, not a Lorenz-system solution.



Rössler-like look-alike

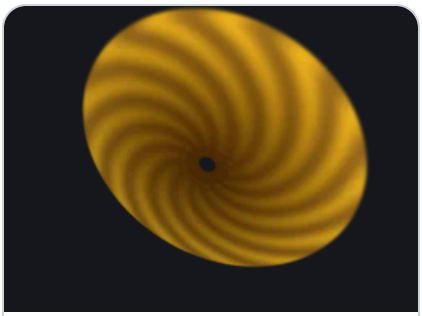
A looping spiral suggests a Rössler attractor but is only a parametric look-alike.



Five-petal flower

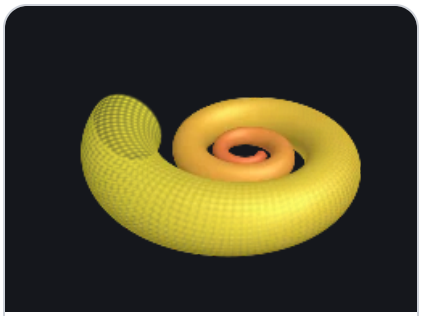
Five broad petals rise from a shallow cup around a small center.

Nature-like shapes (10 programs)



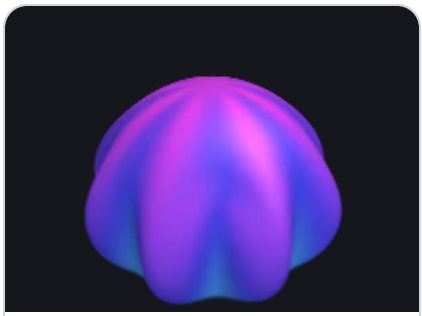
Sunflower-style spiral

A face-on disk shades from dark seeds to a golden rim, with Fibonacci-style spiral arms drawn in the color.



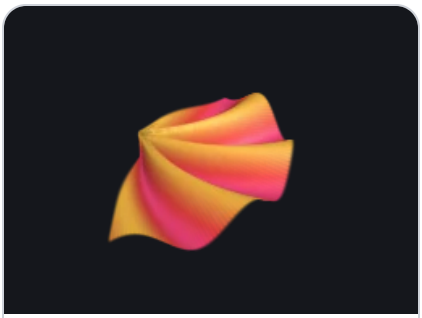
Nautilus coil

A widening spiral tube evokes a nautilus shell with a visible open center.



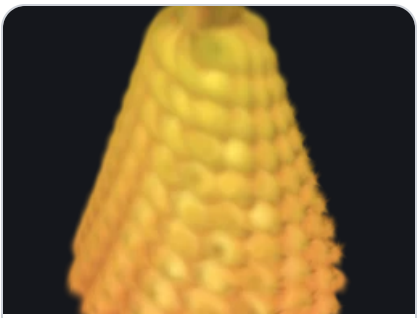
Jellyfish bell

A dome-like bell pulses as its rim widens and narrows once each cycle.



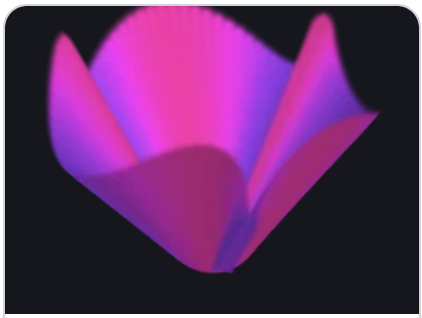
Coral fan

Branching sine ribs spread across a fan-shaped sheet.



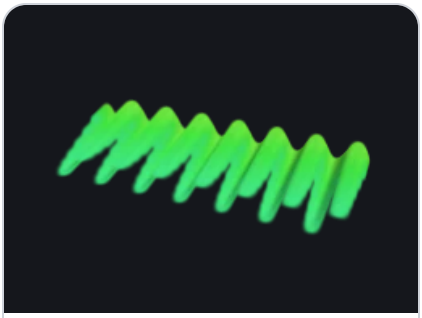
Pinecone spiral

Scales follow a Fibonacci angle and taper toward the top of the cone.



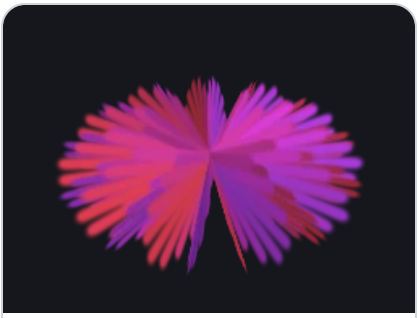
Sea anemone

Curled arms rise from a low base like a ring of waving tentacles.



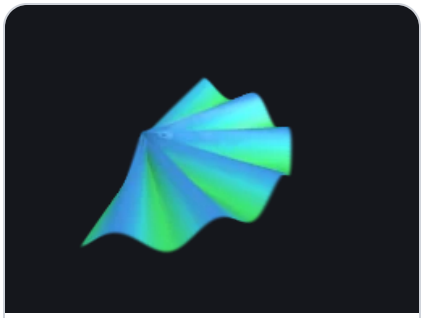
Fern frond

A central stem and repeated leaflets make a simple fern-like frond.



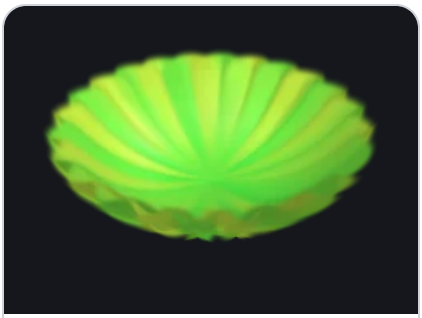
Dahlia bloom

Layered radial petals alternate in height to form a dense dahlia-like flower.



Sea fan branches

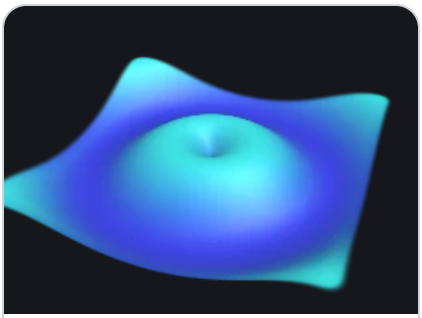
A flat branching fan uses nested waves to suggest coral growth.



Lily pad

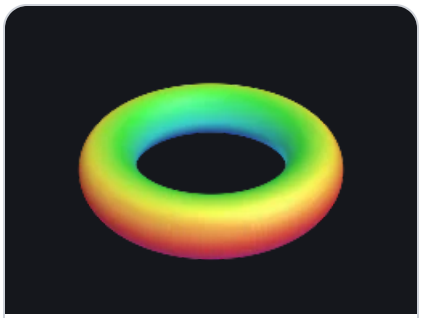
A scalloped circular leaf floats on a softly raised surface.

Things in motion (10 programs)



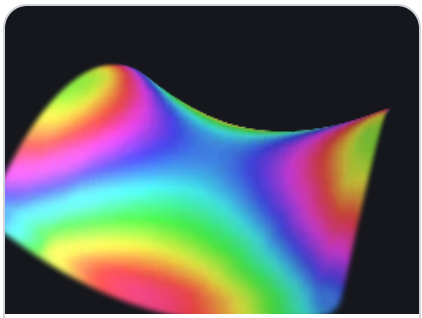
Traveling wave

Concentric ripples travel across a square sheet without breaking at the cycle seam.



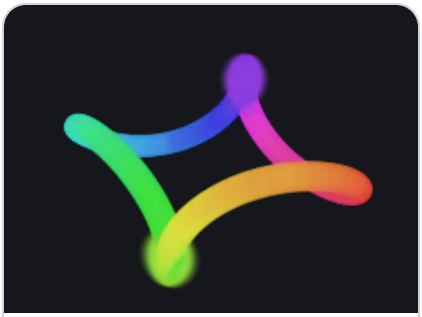
Breathing torus

The major and minor radii expand together and return once per cycle.



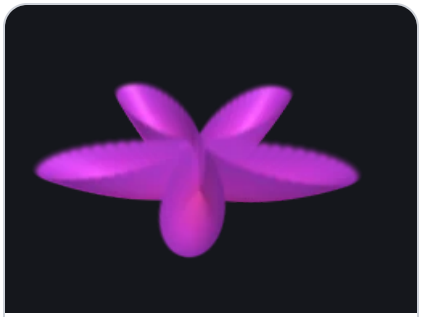
Rotating saddle

A saddle ripple rotates smoothly over a square patch.



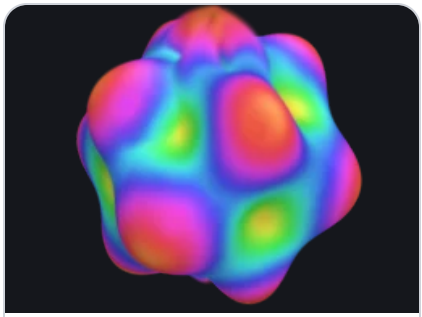
Orbiting rings

Three circular bands revolve at different speeds around the center.



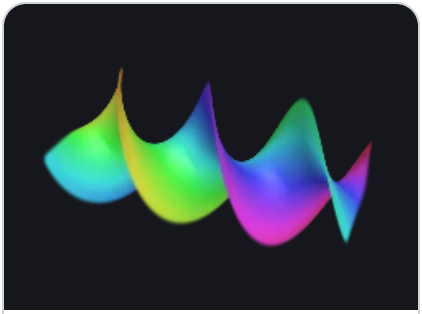
Opening flower

Petals lift and flatten in a smooth repeating bloom.



Breathing bumpy sphere

Four-fold bumps grow and shrink across a round surface as its color follows each bump.



Twisting ribbon

A narrow ribbon rotates its cross-section through one full turn each cycle.



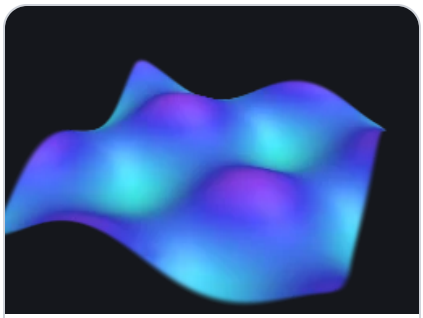
Pulsing star

Five radial points breathe in and out with the beat of a sine wave.



Wobbling helix

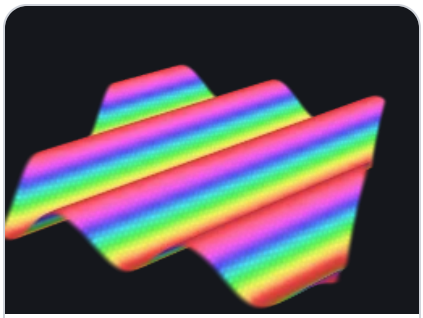
A helix sways side to side and returns to its starting pose each cycle.



Standing wave

A rectangular sheet ripples in place with a standing two-axis wave.

Color play (10 programs)



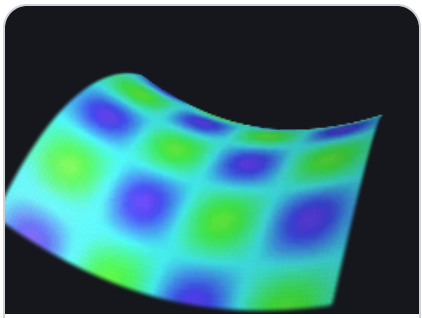
Height hue bands

A ridged surface assigns a full rainbow cycle to its height.



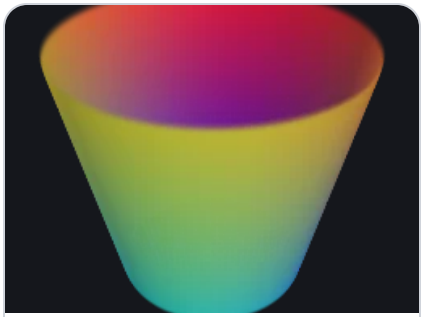
Angle around the torus

Hue sweeps around the large ring while a second color band follows the tube.



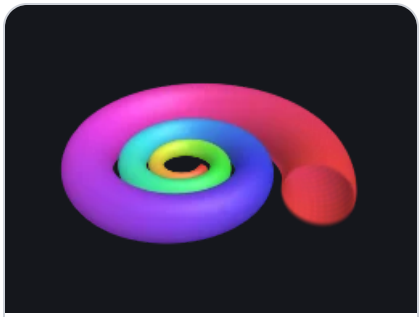
Curvature-like bands

Nested sine ridges color a saddle with a pattern that suggests changing curvature.



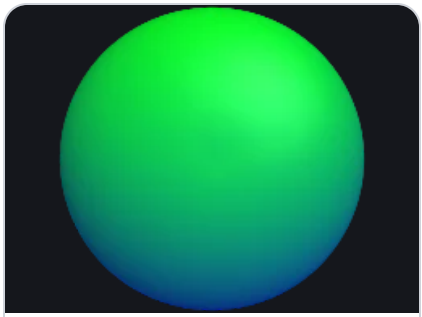
RGB sunset gradient

Red rises with height, green follows the horizontal angle, and blue fills the valleys.



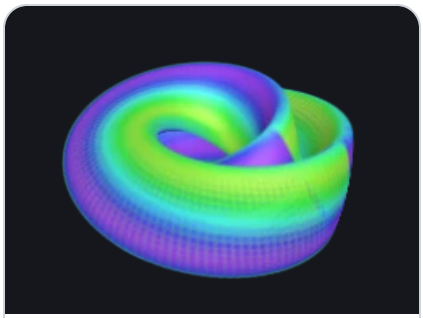
Rainbow shell

A growing shell carries a rainbow gradient from its narrow tip to its wide lip.



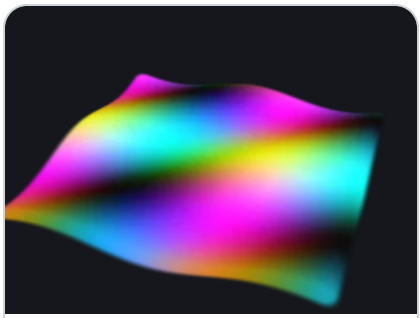
Blue-green sphere

An RGB gradient moves from deep blue at the bottom to green at the top.



Klein bottle color flow

Alternating bands run around the immersed bottle and make its crossing easy to see.



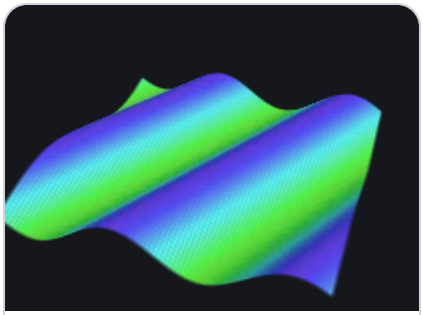
Plasma RGB sheet

Three offset waves make a smooth red, green, and blue color field.



Longitude stripes

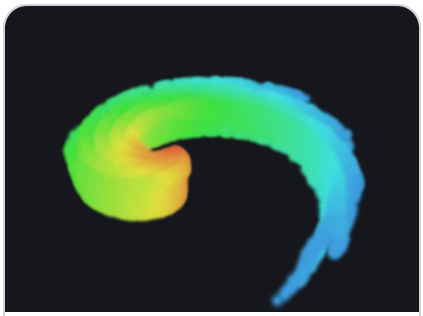
A sphere is divided into twelve clean color wedges by its longitude.



Rainbow wave

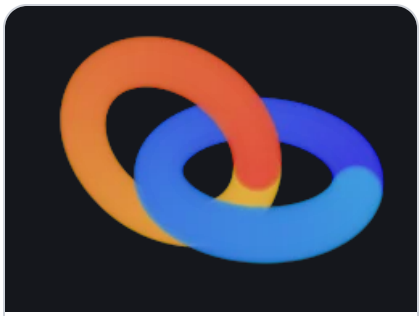
A moving ripple shifts both the sheet height and its rainbow colors.

Math showpieces (10 programs)



Thermal spiral

Warm colors gather near the core and cool colors trail along the spiral arms.



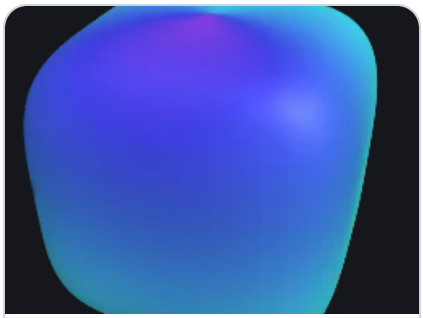
Hopf link

Two circles pass through each other once, the simplest link, drawn as two dotted rings in different colors.



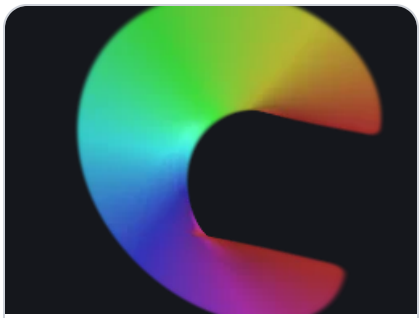
Fourier epicycle curve

Three rotating harmonics add into one closed curve with a small bead radius.



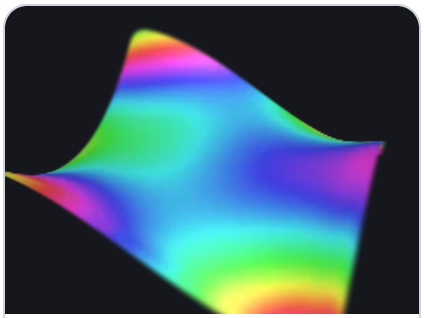
Superquadric

Fourth-power rounding turns a sphere into a soft box with six flattened faces.



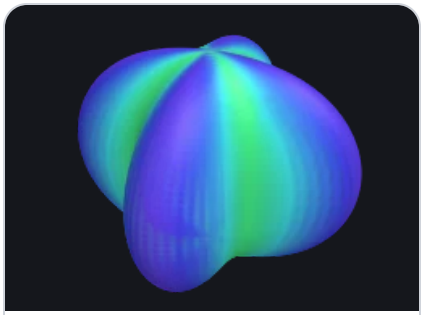
Riemann branch look

A square-root branch surface spirals through two sheets around its branch point.



Monkey saddle

A cubic height field has three valleys meeting at one central saddle point.



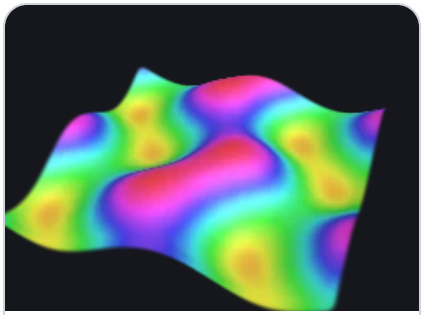
Spherical harmonic lobes

A spherical-harmonic-like radial formula gives the sphere four raised and lowered lobes.



Lemniscate tube

A tube wraps around a sideways figure eight with a subtle vertical wave.



Gyroid slice

A height graph of one slice of the gyroid function rises and falls in a repeating pattern of hills and valleys.



Torus knot family

A (2,3) torus knot winds twice around the ring and three times around its tube.

Things to try. Change a number and see what moves. Swap `cos` and `sin`. Put *t* somewhere new. Make `count` small and `size` big to see the single splats. Try `spread = random` on a surface. Give a sphere a `hue` of `floor(8v/pi)/8`.

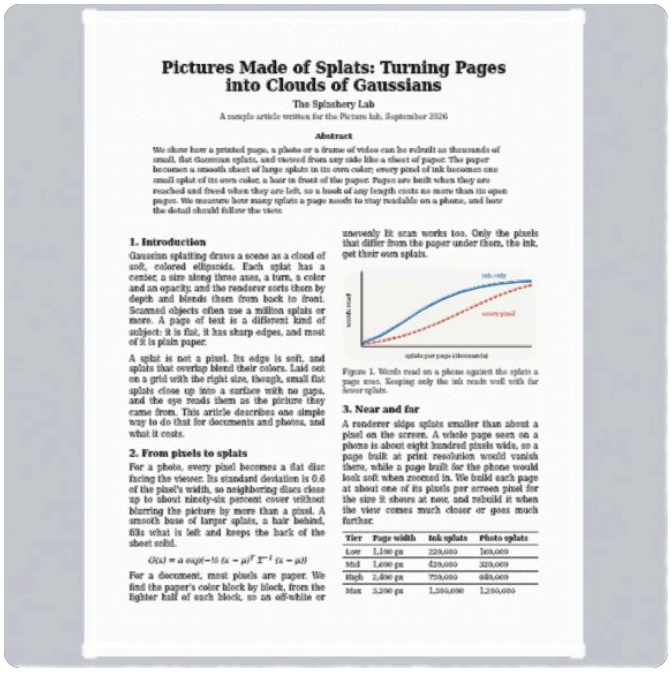
CHAPTER 5

Pictures, screens, and your own input

The toys in Level 2 are built from shapes. Some toys show something that is not a shape: a page of a PDF, a photo, a GIF, a video, or something you type or draw. Three additions to a recipe make them: **picture sheets**, an **input panel**, and a few **view settings**. They are still Gaussian splats, all the way down. The toys on the Pictures and pages shelf are being tried out, so they are on the shelf only when Splashery is opened with `?labs=1`.

Picture sheets: a page, a photo, a GIF, a video

A **sheet** is a flat place in a toy where a picture shows: `k.sheet({ id, center, width, height })`. The kit draws nothing there. The site opens the picture before the build, then turns it into splats on the sheet after the build, in a background worker so the page stays smooth. A PDF page becomes a smooth sheet of large soft splats in the paper's color, with a small splat of its own color wherever the page differs from the paper, which is the ink (`method: "ink"`); in a photo, each sampled pixel becomes a flat splat (`"pixels"`); a GIF or a video becomes a screen that is redrawn on every frame (`"screen"`). The site picks the method from the kind of file. It rebuilds a sheet finer as you zoom in, and frees the pages you have left, so a book of 200 pages does not need 200 pages of splats.



Pictures Made of Splats: Turning Pages into Clouds of Gaussians

The Splashery Lab

A sample article written for the Picture lab, September 2026

Abstract

We show how a printed page, a photo or a frame of video can be rebuilt as thousands of small, flat Gaussian splats, and viewed from any side like a sheet of paper. The paper becomes a smooth sheet of large splats in its own color; every pixel of ink becomes one small splat of its own color, a hair in front of the paper. Pages are built when they are reached and freed when they are left, so a book of any length costs no more than its open pages. We measure how many splats a page needs to stay readable on a phone, and how the detail should follow the view.

1. Introduction

Gaussian splatting draws a scene as a cloud of soft, colored ellipsoids. Each splat has a center, a size along three axes, a turn, a color and an opacity, and the renderer sorts them by depth and blends them from back to front. Scanned objects often use a million splats or more. A page of text is a different kind of subject: it is flat, it has sharp edges, and most of it is plain paper.

A splat is not a pixel. Its edge is soft, and splats that overlap blend their colors. Laid out on a grid with the right size, though, small flat splats close up into a surface with no gaps, and the eye reads them as the picture they came from. This article describes one simple way to do that for documents and photos, and what it costs.

2. From pixels to splats

For a photo, every pixel becomes a flat disc facing the viewer. Its standard deviation is 0.6 of the pixel's width, so neighboring discs close up to about ninety-six percent cover without blurring the picture by more than a pixel. A smooth base of larger splats, a hair behind, fills what is left and keeps the back of the sheet solid.

$$G(x) = a\,\exp\!\left(-\tfrac{1}{2}\,(x-\mu)^{T}\,\Sigma^{-1}\,(x-\mu)\right)$$

For a document, most pixels are paper. We find the paper's color block by block, from the lighter half of each block, so an off-white or unevenly lit scan works too. Only the pixels that differ from the paper under them, the ink, get their own splats.

Figure 1. Words read on a phone against the splats a page uses. Keeping only the ink reads well with far fewer splats.

3. Near and far

A renderer skips splats smaller than about a pixel on the screen. A whole page seen on a phone is about eight hundred pixels wide, so a page built at print resolution would vanish there, while a page built for the phone would look soft when zoomed in. We build each page at about one of its pixels per screen pixel for the size it shows at now, and rebuild it when the view comes much closer or goes much further.

Tier	Page width	Ink splats	Photo splats
Low	1,100 px	320,000	360,000
Mid	1,600 px	420,000	320,000
High	2,400 px	770,000	680,000
Max	3,200 px	1,310,000	1,210,000

A PDF page (the Picture lab)



The recipe below: a photo in a frame

Both pictures are made of splats. Photo: "Tulip field" by DennisM2, CC0. The article is our own sample text.

Three parts of a recipe work together:

- `pictures: { sample, accept }` names what shows until you open your own (the sample's address) and which kinds a visitor may open: `"pdf"`, `"image"`, `"gif"`, `"video"`.
- `drive` may choose the page of each sheet with `out.sheets[id] = { page, visible }`. Pages count from 0, and a page that does not exist shows nothing. A sheet that is not visible is not built unless it requests `ahead: 1`. Without it, every sheet shows the current page.
- `info.data.pictures` is the toy's view of the media: `page`, `count`, `kind`, `name`, `playing`, and the functions `next()`, `prev()`, `go(n)`, and `togglePlay()`.

A whole picture toy, in a few lines. Its `input` gives the Toy tab a button that opens a photo, a GIF, or a video; the rest of the toy is a card and four wooden bars:

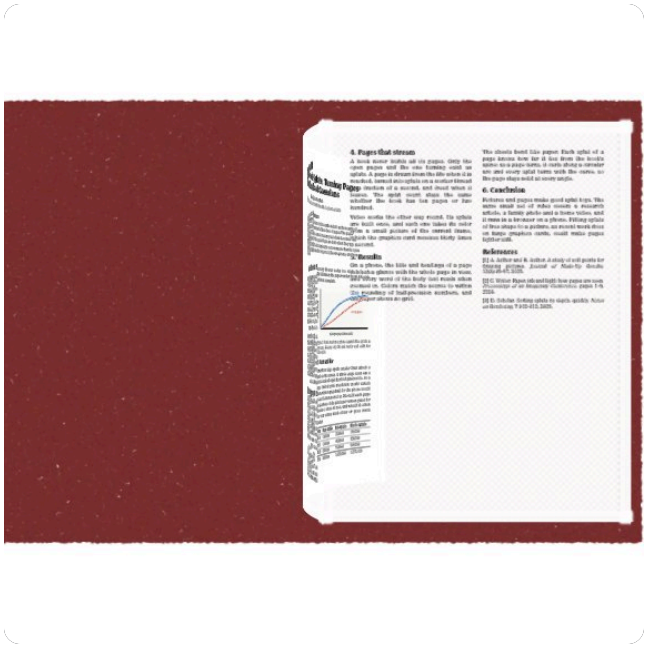
```
import { smoothstep } from "../src/kit.js";

// Two small picture toys for Level 5 of the Tinkerer's Manual: a framed photo
// that opens your own picture, and a little book whose page turns like paper.
export const RECIPES = {
  "little-frame": {
    turntable: false,
    tiltLock: true, // a drag only spins it left and right
    pictures: {
```

```
      sample: () => "assets/toys/picture-lab/photo.jpg", // shown until you open your own
      accept: ["image", "gif", "video"],
    },
    input: {
      title: "Your own picture",
      media: { accept: ["image", "gif", "video"] },
      note: "Open a photo, a GIF or a video. Files stay on this device.",
    },
    build(k) {
      // The picture: a sheet, filled with splats after the build.
      k.sheet({ id: "photo", center: [0, 0, 0.03], width: 1.5, height: 1 });
      // A gray card behind it, and four wooden bars around it.
      k.add(k.box(1.6, 1.1, 0.02), { pos: [0, 0, -0.02], color: "#9a9a9a" });
      for (const [w, h, x, y] of [
        [1.8, 0.1, 0, 0.6],
        [1.8, 0.1, 0, -0.6],
        [0.1, 1.1, -0.85, 0],
        [0.1, 1.1, 0.85, 0],
      ])
        k.add(k.box(w, h, 0.1), { pos: [x, y, 0], color: "#8a5a34" });
    },
  },
```

Turning a page

A page that turns like paper is a sheet with `leaf: slot` (a slot from 0 to 9) and a spine to turn about: `k.spine({ at, axis, dir })` gives the spine's place, its direction, and the direction from it along the pages when they lie flat. Each frame, `drive` sets `out.leaves[slot] = { angle, curl }`: the `angle` in radians (0 lying flat, π turned all the way over) and the `curl`, how much the page bends along a circular arc (a negative number lets the free edge lag behind, like paper in air). Every splat of the page turns with the curve at its place, so it stays a solid sheet at every angle, which a page bent with soft regions would not.



The little book below, halfway through a turn. The next page is already there under the turning one.

The little book below uses a pulse control for the turn, the way Level 2's gust does. The page under the turning one is built ahead of time (a sheet is normally built only while it is visible, unless it requests `ahead: 1`, so a page that shows up halfway could make the turn wait), and the page number moves on when the turn ends. Keep a turning sheet on the body (part 0), because a part's own turn would replace the leaf's. The far side of the turning page shows the page in mirror image, like tracing paper, until it lands.

```
"little-book": {
  turntable: false,
  tiltLock: true,
  pictures: { sample: () => "assets/toys/picture-lab/article.pdf", accept: ["pdf"] },
  input: { title: "Your own book", media: { accept: ["pdf"] }, note: "Open a PDF." },
  controls: [{ key: "turn", label: "Turn the page", type: "pulse", ease: 1.2 }],
  action: { key: "turn", label: "Turn the page" },
  drive(t, c, out, info) {
    const pics = info.data?.pictures; // { page, count, go(n), next(), prev(), ... }
    // The page changes when the turn ends, and the turning page lands on the left.
    const turning = c.turn > 0;
    if (BOOK.turning && !turning && pics) pics.go(pics.page + 1);
    BOOK.turning = turning;
    const page = pics?.page ?? 0;
    // The pulse falls from 1 to 0: the page swings from 0 to a half turn (pi)
    // about the spine, and its free edge lags behind, like paper.
    const s = turning ? smoothstep(0, 1, 1 - c.turn) : 0;
    out.leaves = [{ angle: Math.PI * s, curl: -1.2 * Math.sin(Math.PI * s) }];
    out.sheets = {
      // Once the turn starts, the left page is the one that is about to land on it.
      left: { page: turning ? page : page - 1 }, // a page out of range shows nothing
      under: { page: page + 1 }, // built ahead, and uncovered as the page lifts
      front: { page }, // the turning page (its far side shows in mirror image)
    };
```

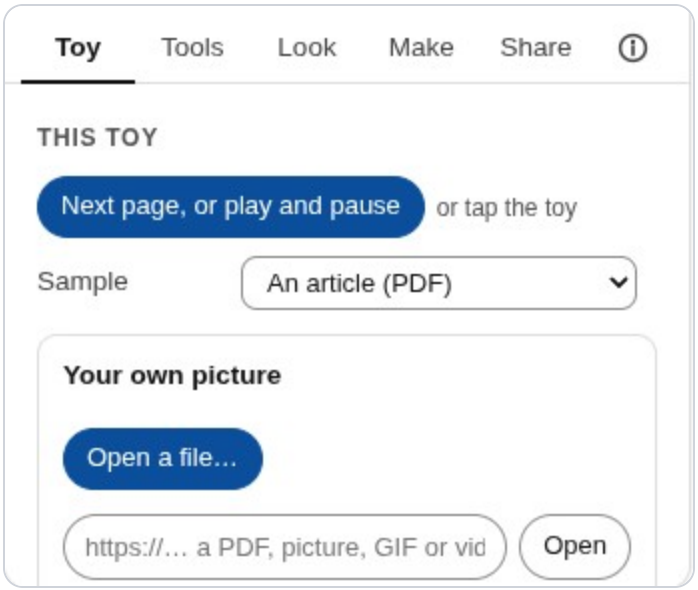
```
      // It is built, but hidden, until the turning page lands on it.
      out.parts.leftPage = { visible: turning ? 0 : 1 };
    },
    build(k) {
      BOOK.turning = false;
      k.spine({ at: [0, 0, 0], axis: [0, 1, 0], dir: [1, 0, 0] });
      const W = 1.4;
      const H = 1.9;
      const leftPage = k.part("leftPage");
      const page = { width: W, height: H };
      k.sheet({ id: "left", ...page, part: leftPage, center: [-W / 2, 0, 0], align: [1, 0] });
      k.sheet({ id: "under", ...page, center: [W / 2, 0, 0], align: [-1, 0] });
      // The turning page: a sheet on leaf slot 0, a hair in front of the others.
      k.sheet({ id: "front", ...page, center: [W / 2, 0, 0.008], align: [-1, 0], leaf: 0 });
      // The cover, behind the pages.
      k.add(k.box(2 * W + 0.1, H + 0.1, 0.03), { pos: [0, 0, -0.03], color: "#7a2f2f" });
    },
  },
};

// Whether the page was turning in the last frame.
const BOOK = { turning: false };
```

Your own input: typing, drawing, opening a file

A recipe can ask for something from the visitor. Its `input` adds a panel to the Toy tab, in one of three forms:

- **Typing**: a text box and a button, and maybe a button that opens a text file (`fileButton` , `accept`). The Splat equation toy of Level 3 is one; so is the Word vectors toy, where you type *king − man + woman*.
- **Drawing**: `pad: { cols, rows, max, button }` adds a small grid you paint with a soft brush, and a button that hands the drawing to `read` as the text `"pad:…"` (one number for each cell, row by row). The CNN toy's "3D, draw a digit" view reads the digit you draw.
- **Opening a picture**: `media: { accept }` adds an **Open a file…** button, a field for a web address, the page buttons (or play and pause for a video) and a way back to the sample. Level 5's picture toys use it.



The input panel of a picture toy, in the Toy tab. Files are read on your device; nothing is uploaded.

For typing and drawing, `read(text, fileName)` turns what came in into option values, or throws an `Error` whose message the panel shows in place; and `shown()` says what is showing now. Keep what goes into options small: options are saved in the toy's link. Here is a whole toy with a typed input, a row of bars for the numbers you give it:

```
// A small toy with its own typed input, for Level 5 of the Tinkerer's Manual:
// type some numbers and the toy stands them up as bars.
const LIMIT = 12;
const COLORS = ["#e8384f", "#ffc233", "#1fb59a", "#4dabf7"];

export const RECIPES = {
  "little-bars": {
    options: [
      {
        key: "numbers",
        label: "Your numbers",
        type: "text",
        default: "3 1 4 1 5 9 2 6",
        hidden: true,
      },
    ],
    input: {
      title: "Your own numbers",
      placeholder: "3 1 4 1 5 9 2 6",
      button: "Show them",
      fileButton: "Open a text file…",
      accept: ".txt,.csv",
      note: `Type up to ${LIMIT} numbers from 0 to 20, separated by spaces or commas, or open a text
file with them.`,
      // Turns the typed text into option values, or throws a message that the panel shows.
      async read(text) {
        const list = String(text).match(/-?\d+(\.\d+)?/g) || [];
        if (list.length === 0) throw new Error("Type some numbers, like 3 1 4 1 5.");
        if (list.length > LIMIT) throw new Error(`That is more than ${LIMIT} numbers.`);
```

```
        if (list.some((n) => n < 0 || n > 20)) throw new Error("Use numbers from 0 to 20.");
        return { numbers: list.join(" ") };
      },
      shown: () => "Your numbers",
    },
    build(k, o) {
      const values = String(o.numbers).split(" ").map(Number);
      values.forEach((v, i) => {
        const h = 0.05 + v / 10;
        k.add(k.box(0.14, h, 0.14), {
          pos: [(i - (values.length - 1) / 2) * 0.2, h / 2, 0],
          color: COLORS[i % COLORS.length],
        });
      });
    },
  },
};
```

The Screen and the Gaussian splatting toy

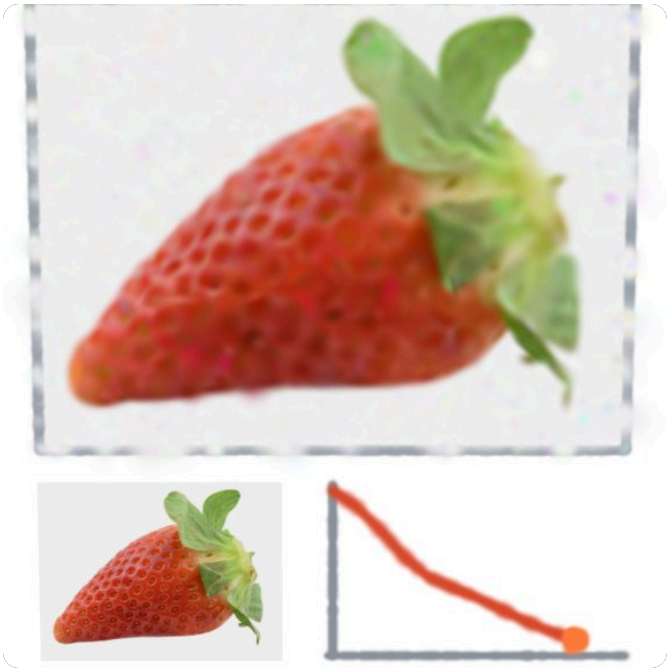
Two more toys put these to work: the Screen on the Pictures and pages shelf, and Gaussian splatting on the AI and computing shelf.

The **Screen** shows a video or a GIF on a screen in one of four styles: an old TV, a flat TV, a cinema with curtains, and a hologram. Its picture is one sheet with `method: "screen"`; the knobs, the curtains, and the glass are ordinary kit parts. A tap switches it on, and then plays and pauses; the speaker button in the top bar turns the sound on. You can open your own video or GIF.

The **Gaussian splatting** toy lets you watch how a photo becomes splats. Its **Training** view fits 2,400 flat (two-dimensional) Gaussians to a photo (a strawberry, or your own), by gradient descent in a background worker: the splats start as random specks and, step by step, they slide, stretch, and change color until together they are the picture. The fit uses front-to-back blending, equivalent to the renderer's back-to-front blending, with the renderer's finite bell, so what you see is the real fit. **One splat** lets you set the sizes, the opacity, and the color of a Gaussian illustrated by a cloud of small splats and turn it (Level 1's numbers); **Many splats** builds a small rubber duck from kit shapes and then shrinks every splat to a dot, so you see the splats that make it; and **Sorting** shows the splats of a ball appearing one by one, in the back-to-front order a camera draws them.

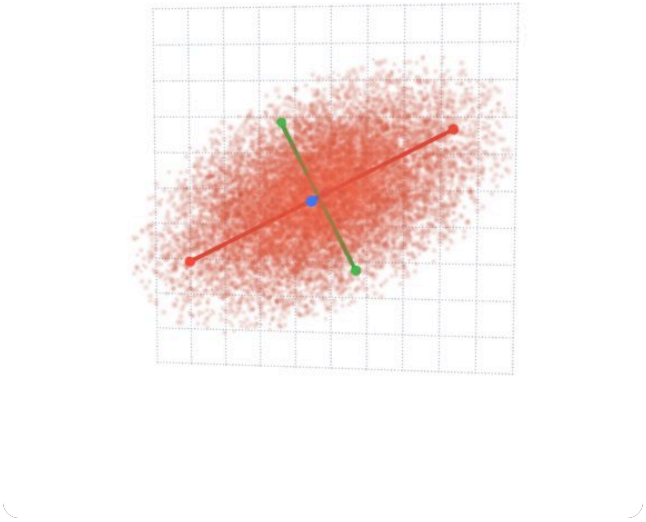


The Screen (old TV)



Gaussian splatting, training

Video frame from "Big Buck Bunny" © Blender Foundation, CC BY 3.0. Strawberry photo by Joselodos, CC0.



The One splat view: a Gaussian illustrated by small splats, with its three sizes drawn as axes (Level 1's s_1, s_2, and s_3).

Steering the view: the top bar and the tilt lock

Three round buttons sit in the top bar, beside the sound and help buttons. They control how you look at the toy on show, and none of them changes the toy.



- **Reset view** (the circular arrow) puts the camera back where the toy starts. The **R** key does the same, and so does the button in the Tools tab. A double tap resets the view unless the toy uses it to focus on a page.
- **Tilt lock** (the padlock) makes a drag turn a toy only left and right, so a flat toy such as a page, a photo, or a screen stays facing you. Zooming, panning, and pinching still work. Your choice is remembered for that toy for the rest of your visit. In a recipe, it is the flag `tiltLock: true`, and every toy on the Pictures and pages shelf sets it. A link keeps the camera it was saved with, and the lock then holds that tilt.
- **Turntable** (the record player) switches the slow turn that a toy does when nobody is touching it. It is the same setting as **Turntable when idle** in the Toy tab, and this device remembers it. A recipe can opt out with `turntable: false`, as a page does. If your device asks for less motion, the turn stays off and the button is disabled.

Two fingers on a touch screen zoom. They roll the toy only after a clear twist (about 17 degrees) and turn it only when they move together, so a normal pinch never spins what you are looking at. A two-finger drag moves the toy across the screen instead of turning it, as do Shift or Option/Alt and a drag on a computer; ↺ or R puts it back in the middle. A double-tap does too unless the toy uses it for page focus.

CHAPTER 6

Moving things

This level will cover hands-on play, joints, and physics. **Coming in the next round.**

CHAPTER 7

Real data as splats

This level will cover science, imaging, maps, and the night sky. **Coming in the next round.**

CHAPTER 8

Photos and video into 3D

This level will cover depth, the Sharp view, and splats. **Coming in the next round.**

CHAPTER 9

Codes and pages

This level will cover QR codes, barcodes, PDFs, and books. **Coming in the next round.**

CHAPTER 10

Make your own

This level will cover Code a toy, the Studio, and the program gallery. **Coming in the next round.**

What else Splashery does now

This chapter lists what is on the site today, beyond the equation toy and picture toys you have read about. It describes only what is built. The **labs switch** hides newer parts until they are ready: add `?labs=1` to the site's address to turn it on (the browser remembers it; `?labs=0` turns it off). Each paragraph says what is behind it. The same list is a page on the site: What Splashery can do.

Hands-on physics

With the ✋ switch on, a drag picks up whatever is under your finger and a quick let-go tosses it. It lands, bounces, and settles on the floor under the toy, inside an invisible play area, and soft toys squish when they land. The ↺ button sends everything home. Every toy plays this way, scans included, as one rigid body; a recipe's `hands` block can instead make its pieces separate bodies, so a toy can have its own interactions. This is not behind the labs switch.

The QR code toy

The QR code toy makes a code from a link, plain text, a Wi-Fi network, a contact card, or an email, and builds it from splats. Its motions (assemble, flip, burst, and return, and an "alive" color wave) move each piece as a solid tile, and the alive wave keeps the gray values a reader sees unchanged, so each frame still scans. A "Scan view" turns the code flat to the camera, and after every change the toy renders that view and reads it back, to check that it scans. The reader is the browser's own detector where there is one, and otherwise the vendored jsQR library; nothing leaves the device. A Wi-Fi password is not kept in a link. This toy is behind the labs switch.

Captures and file formats

The viewer opens `.ply`, `.sog`, `.splat`, and `.spz` files (version 4 SPZ files, which need a decoder Splashery does not ship, are refused with a message). Files can carry spherical harmonics, the view-dependent color the kit's splats do not use. The Scans shelf holds captures with their licenses and credits. The newer captures each ship as a full file and a lite file, and some keep a band of spherical harmonics on the full file. Many of the newer captures are behind the labs switch.

Camera, microphone, and screen

A toy can read the microphone, the camera, or a shared screen, but only after you tap a button for it. Nothing is requested before that tap. While a source is on, a small "Live" pill with a Stop button shows at the top of the page, and the sound or pictures stay on your device: nothing is recorded or sent. Two examples are the room echo meter (clap, and it measures how long the room rings) and the splat mirror (your camera as splats, with a depth model of about 27 MB that loads only after you start the camera). Both are behind the labs switch.

Books, albums, and frames

Your book turns the pages of a PDF as paper: a page is a leaf with a front and a back, the engine can re-sort the splats of a turning page where it stands, and a turn waits until the next page is ready so it never shows a blank one. The words of a PDF page are available from its text layer, and links in the PDF are tinted on the page. The photo album opens several photos at once, and the picture frame lets you list them and move each up or down. These toys are behind the labs switch.

Studio: sound and photos

The Studio shelf turns other things into splats on your device. The song landscape makes a spectrogram of a song you open into a 3D field, and the Chladni plate shows sand finding a plate's still lines. Photo to 3D runs a depth model on your device, works out how far each part of a photo is, and rebuilds the photo as splats at their depths. The Studio toys are behind the labs switch.

Models to splats

The Model to splats toy opens a glTF or GLB file, an OBJ with its MTL, or an STL file and turns it into splats on your device: flat discs on the surface, more of them where there is detail, each sized to its neighbors so the surface closes, and colored from the model's texture, vertex colors, or material. A toy's input can take several files at once, for a glTF with its `.bin` and textures. This toy is behind the labs switch.

Worlds

A world is a small place you walk around in, built on your device from recipes and a seed. It has a start screen, a list of places, collision, a character, and a camera that follows. In splats mode everything you see is splats. In hybrid mode the ground, water, sky, and signs are lit models, and only the props and the near grass stay splats, so "everything is splats" holds for the toys but not for every part of every world. Worlds are behind the labs switch (`worlds/?labs=1`).

Fluids, songs, and smooth bodies

A recipe can declare a liquid, a gas or a flame with `k.fluid`. It is simulated while the toy is open and drawn as its own splats, one per particle, and the engine loads only when such a toy opens. A keyboard toy can play songs from its own list, a MIDI file, or a tune pasted as ABC notation, with every key, hammer, and string moving as a solid piece. Recipes can also join shapes into one smooth body and place one toy's recipe inside another's (the anatomy atlas does both). The Fluid lab, the bottle toys, and the anatomy atlas are behind the labs switch.

Solid, Fine, and Dots, and the detail you see

The Splat equation toy has three looks. **Solid** lays a surface's splats flat along it so they close into a clean shape, **Fine** uses up to four times as many smaller splats within the device's budget, and **Dots** shows each splat you programmed as its own round dot. The toy also asks for the sharper labs kernel. Beyond that toy, the viewer's Detail setting (Auto, High, or Max, and with labs on a slider from 60,000 to 400,000 splats) lives in your browser only, never in a link, and on every toy the canvas resolution drops while you drag if frames are slow. Those two are not behind the labs switch; the toy and the sharper kernel are.

The viewer's controls

Focus mode hides everything but the toy (the **F** key or the focus button, and **Esc** brings the rest back). On a computer, **[** folds the side panel and **G** opens the gallery. A toy's input panel can take a typed text, a drawing, a file, a binary file such as a sound, or several files at once. These controls are for everyone.

What Splashery can do that other splat tools can't

PlayCanvas is the engine that draws the splats, and its editor, SuperSplat, works on captured splat files: you crop, clean, and arrange a capture. Splashery starts from the other end, because most of its toys are never captured at all. These are things we have not seen in other public splat tools. (We have not tried every tool, so read this as "to our knowledge".) The site has this list as a page too: What Splashery can do.

- **Toys made from rules, in the browser.** A recipe of a few kilobytes becomes 60,000 to 400,000 splats on your device, depending on the device and the Detail setting. There is no multi-megabyte file to download.
- **Equations as a language.** Type *x*, *y*, and *z* in *u*, *v*, and *t*, and get an animated splat shape, with plain messages instead of crashes (Level 3).
- **Moving parts inside one splat cloud.** Hinges, pivots, sliders, morphs, and per-splat effects run on the graphics chip, so a chest opens and a windmill turns without rebuilding anything (Level 2).
- **Hands-on physics on splats.** Toss, stack, pull apart, and pour.
- **Anything as splats.** Photos and videos with depth, PDFs, QR codes, real science data, the night sky, and medical volumes.
- **A whole scene in a link.** The `#s=` links carry the toys, their options, and the camera. How they work:
  - The scene is compressed with deflate and then written in base64url (`src/codec.js`). Where a browser cannot compress, the link carries the plain text instead and says so.
  - A link carries settings, not your files. A photo or a splat file from your device has to be opened again by whoever opens the link.
  - No server is involved. The link is the whole scene.

Open questions: other choices, not tried yet

Splashery works as it is, and nothing here would replace what works. Each of these would be a separate experiment, behind the labs switch or on a comparison page, with today's behavior as the default. If you wonder "why this number?", the answer is in the notes in Levels 1 to 3, or it is "not tested yet" and listed here.

Show the table: every choice not tried yet

Choice	What an experiment would show	Status
A Detail slider	Any splat count from 60,000 to 400,000, beside the tiers. Built as a labs slider (`docs/lab/DETAIL-SLIDER.md`).	Built, in labs
View-dependent color	A shine that moves with the view on a few kit toys, from the kit's own data and shader. Built as a private comparison, on only when a link or the address asks for it with labs on (`docs/lab/GLOSS.md`).	Built, in labs
More moments of *t*	Turning shapes at more than 12 moments, up to 15. Built as a labs choice (`docs/lab/MOMENTS.md`); more than 15 is still open.	Built, in labs
240-character fields	Longer fields, and what longer links cost. Built in the Splat equation toy (`docs/lab/EQUATION-FIELDS.md`).	Built, in labs
A budget sweep on real phones	Real phone graphics chips, and more phone sizes. The Kit lab's sweep (Level 2) covered five budgets, one phone size, and one software renderer.	Not started
More than 15 parts	A toy with 24 or 32 moving parts. The largest change: it repacks the part number, moves the part table into a texture, and needs a check on real phones.	Open: no plan yet
An opacity field	See-through splats in a program (and the other candidate fields in Level 3).	Open: no plan yet
Formulas on the graphics chip	Formulas compiled to a shader, so *t* is continuous and no copies are stored.	Open: no plan yet

What comes later: Code a toy

The splat equation language can place and color splats, but it can't give a toy parts, controls, or a tap of its own. For that you need a full recipe (Level 2). Splashery's plan includes a **Toy Workshop** where you will be able to write one: a **Code a toy** editor that builds and shows your recipe as you type.

Sharing those toys will work differently from sharing equations. A recipe is real JavaScript, and a page that ran a stranger's JavaScript straight from a link could be made to do things its visitor never agreed to. So a coded toy will be shared as a **submission**: it is checked and read by a person before it joins the shelf, the way every toy in Splashery is today. Equations stay the quick way to share; recipes are the deep one.

Until then, everything in Level 2 is in the source code, which is open under the MIT license: github.com/ryanjosephkamp/splashery.

Glossary

Gaussian (peak-one bell)

A smooth weight that falls off exponentially with the squared distance from a center, measured in units of the widths. In this manual it peaks at 1.

Weight field, positive widths

A weight field is a number at every point in space: 1 at the center, falling off with distance. The three widths are a splat's sizes along its own axes; they must be above zero.

Mean, center μ

The position where the bell has its largest weight.

Standard deviation, axis width s

A length that measures distance along one of a splat's own axes. It is a yardstick, not a hard radius: the ideal weight is still about 0.14 at two widths.

Covariance Σ

A symmetric 3-by-3 matrix that holds the squared widths and the way the axes are turned, $\Sigma = R\,S\,S^{\mathrm{T}}\,R^{\mathrm{T}}$. Terms off the diagonal mean the axes are tilted.

Quadratic form

An expression in which every term has two factors of the input. In two dimensions, $ad_1^2 + 2bd_1d_2 + cd_2^2$. With 4 and 1 on the diagonal it is $4d_1^2 + d_2^2$, and setting it to 1 draws an ellipse. The exponent of a Gaussian holds one [5].

Tilt, rotation

Tilt is the off-diagonal terms of Σ: zero when a splat's axes line up with x, y, and z. Rotation is a splat's own turn. "Tilt lock" is a different thing: the camera control in Level 5.

Anisotropic, isotropic

Different widths along the axes, or equal widths in every direction. A plain kit splat is isotropic; the flat discs that cover a surface are anisotropic.

Quaternion q

Four numbers that describe a rotation. They only describe a rotation when their length is 1, so a reader scales them to length 1 first.

Affine

A straight-line map plus a shift, $p \mapsto Ap + b$. Straight lines stay straight [6].

Footprint

The patch of the screen where one splat can add any color.

Jacobian *J*

The matrix of the projection's slopes at one point, used as a straight-line stand-in for the projection near the splat's center.

Opacity *α*, transmittance *T*

Opacity says how much a splat covers. Transmittance is how much of what lies behind still shows after the splats in front: the product of $(1 - a)$ over them.

Alpha compositing

Laying colors over one another in order, each weighted by its opacity and by how much of the pixel is still uncovered.

Spherical harmonics

A compact way to store how a color changes with the direction you look from. Many trained captures store them; a kit splat has one base color.

Recipe, capture

A recipe is a program that makes splats. A capture is a file of splats fitted to photographs of a real thing.

Kit, kit toy

The kit is Splashery's toy-building library. A recipe names shapes, colors, and moving parts, and the kit fills them with splats in the browser, spreading the budget evenly by area. A kit toy is built this way; a scan or capture is loaded from a file.

Smoothstep

The curve $3g^2 - 2g^3$, an S-shaped ramp from 0 to 1 that is flat at both ends, used to start and stop a motion gently [7].

Grammar, atom

A grammar is a list of rules for what counts as a valid program. In it, `=` is "is made of", `|` is "or", `{ }` repeats, `[ ]` is optional, and quotes mark text typed exactly. An atom is the smallest piece of an expression that can stand alone [8].

u, v, t, morph

u and *v* are the coordinates of the flat sheet of points that each splat gets a place on, before the equations bend it into shape. They are parameters of a parametric surface, not a substitution [9]. *t* is time: 0 to 2π in the equation language, and the toy's running time in a recipe's `drive`. A morph is a splat sliding from its own place to a target place.

Kernel

The rule for how a splat's weight falls off across its patch. The site normally draws a cut-off Gaussian; a labs setting can use a sharper one. Neither is the ideal bell.

Further reading

Five sources, each opened, and checked on October 4, 2026. Authors, titles, venues, and years are as the source's own page gives them.

- Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. **3D Gaussian Splatting for Real-Time Radiance Field Rendering.** SIGGRAPH 2023 (ACM Transactions on Graphics). Project page · Paper. The starting point.
- Matthias Zwicker, Hanspeter Pfister, Jeroen van Baar, and Markus Gross. **EWA Volume Splatting.** Proceedings of IEEE Visualization 2001. Paper. The footprint step in Level 1.
- Zehao Yu, Anpei Chen, Binbin Huang, Torsten Sattler, and Andreas Geiger. **Mip-Splatting: Alias-free 3D Gaussian Splatting.** CVPR 2024 (Oral, Best Student Paper). Project page.
- Binbin Huang, Zehao Yu, Anpei Chen, Andreas Geiger, and Shenghua Gao. **2D Gaussian Splatting for Geometrically Accurate Radiance Fields.** SIGGRAPH 2024. Project page. A different method from Splashery's flat one-photo fit.
- Antoine Guédon and Vincent Lepetit. **SuGaR: Surface-Aligned Gaussian Splatting for Efficient 3D Mesh Reconstruction and High-Quality Mesh Rendering.** CVPR 2024. Project page.

References

Numbered sources, cited in the text by their number in square brackets. Each entry gives the authors, the title, where it appeared, the year, and a link. "Further reading" above lists more papers on splats.

1. Kerbl, B., Kopanas, G., Leimkühler, T., and Drettakis, G. "3D Gaussian Splatting for Real-Time Radiance Field Rendering." *ACM Transactions on Graphics* (SIGGRAPH), 2023. arxiv.org/pdf/2308.04079
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8. "Extended Backus–Naur form." *Wikipedia*, the free encyclopedia. en.wikipedia.org/wiki/Extended_Backus–Naur_form
9. "Parametric surface." *Wikipedia*, the free encyclopedia. en.wikipedia.org/wiki/Parametric_surface
10. Kamp, R. J. "Writing toys for Splashery packs" (`docs/PACKS.md`). *Splashery* source code, 2026. github.com/ryanjosephkamp/splashery/blob/main/docs/PACKS.md

Quick reference

One splat

$$G(p) = \exp(-\tfrac{1}{2}\,(p-\mu)^{\mathrm{T}}\,\Sigma^{-1}\,(p-\mu)), \quad \Sigma = R\,S\,S^{\mathrm{T}}\,R^{\mathrm{T}}$$

$$\Sigma' = J\,W\,\Sigma\,W^{\mathrm{T}}\,J^{\mathrm{T}}, \quad C \leftarrow a\,c + (1-a)\,C \text{ (back to front)}$$

A splat program

```
x = …        y = …        z = …        # in u, v and t (y is up)
u = 0 .. 2pi                           # ranges: numbers only
v = 0 .. pi                            # default 0 .. 1
hue = …                                # 0 red, 1/3 green, 2/3 blue
r = …   g = …   b = …                  # or colors from 0 to 1
size = 0.05                            # about the bell's σ
count = 4000                           # 100 to 10,000
spread = grid                          # or random
```

Functions: sin, cos, tan, asin (arcsin), acos (arccos), atan (arctan), sinh, cosh, tanh, exp, log (ln), log10, sqrt, abs, floor, min, max. **Constants**: pi, e. **Time**: *t* runs from 0 to 2π per tap. At most 240 characters a field.

A picture toy

```
pictures: { sample: () => "assets/…", accept: ["pdf", "image", "gif", "video"] },
input: { title, media: { accept } },     // or placeholder, button, read, shown; or pad
tiltLock: true,                          // a drag turns it only left and right
build(k) { k.sheet({ id, center, width, height, leaf }); k.spine({ at, axis, dir }); },
drive(t, c, out, info) {
  out.sheets[id] = { page, visible };    // pages count from 0
  out.leaves[slot] = { angle, curl };    // radians; 0 flat, pi turned over
  info.data.pictures                     // page, count, go(n), next(), prev(), togglePlay()
}
```

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